

THE WEAK NEUTRAL CURRENT AND ITS EFFECTS IN STELLAR COLLAPSE

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Daniel Z. Freedman

Institute for Theoretical Physics, State University of New York at Stony Brook,
 Stony Brook, New York 11790

David N. Schramm¹ and David L. Tubbs²

Enrico Fermi Institute (LASR), University of Chicago, Chicago, Illinois 60637

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1 INTRODUCTION

In the summer of 1973, experimental results indicating the presence of a neutral current in the weak interactions were announced, and the effect has since been confirmed in all other laboratories with neutrino-beam facilities (1-7). The conventional charged-current description of weak interactions goes back to Fermi (8), and the modern version evolved rapidly after the discovery of parity violation in β decay (9-12), the formulation of the conserved-vector current hypothesis (13-15),

¹ Also Department of Astronomy and Astrophysics.

² Also Department of Physics.

and the discovery that there were two distinct neutrino species, ν_e and ν_μ (plus corresponding antineutrinos), associated with the e^- and μ^- particles (16). The charged current requires that any particle reaction initiated in a ν_μ neutrino beam, such as $\nu_\mu + n \rightarrow \mu^- + p$, must have a μ^- particle in the final state. The existence of a neutral current was inferred because of the observation of reactions whose final state did not contain the required charged lepton. Examples of neutral-current reactions that have been observed directly are $\nu_\mu + e \rightarrow \nu_\mu + e$ (1), $\nu_\mu + p \rightarrow \nu_\mu + p + \pi^0$ (5, 17), and $\nu_\mu + p \rightarrow \nu_\mu + p$ (18, 19). The final-state neutrino cannot be detected, but its presence is confirmed by kinematics, i.e. energy and momentum conservation. Originally, the existence of the neutrino was inferred from similar kinematic arguments in nuclear β decay.

The neutral current of the 1973–1974 experiments was not born into an unprepared physics world, because it had been predicted in theoretical work by Weinberg in 1967 (20) and Salam in 1968 (21). In this work, the neutral current is a natural feature of a unified treatment of weak and electromagnetic interactions that is based on non-Abelian gauge-field theory of a type first invented by Yang & Mills (22), together with a spontaneous symmetry-breaking mechanism studied by Higgs and others (23–25). An extraordinary synthesis of theoretical ideas is contained in the work of Weinberg and Salam, whose general correctness is confirmed by the experimental discovery. We refer readers to (26–29) for recent reviews of the theoretical framework.

It has been realized for many years (30) that neutrinos play an important role in stellar evolution. This occurs simply because their interaction with matter is so weak that under most circumstances a neutrino escapes instantaneously from a stellar interior, carrying away energy and momentum. As we discuss later, this energy-loss mechanism determines both the time scale and the qualitative features of the intermediate and late stages of stellar evolution (31).

Transparency of matter to neutrinos is not an absolute concept; it is a quantitative question of the size of the neutrino mean free path λ relative to the radius R of the system. Matter is transparent for $\lambda \gg R$ and opaque for $\lambda \ll R$. The mean free path depends on the neutrino energy and composition, density, and temperature of the material. For a 20-MeV ν_e or $\bar{\nu}_e$ in matter consisting of protons and neutrons not bound in nuclei, plus electrons, the cross section is 10^{-41} cm², and one finds $\lambda \approx 10^{17} \rho^{-1}$ cm. In the hydrogen-burning stage of an ordinary star, we have $\rho \sim 1$ gm cm⁻³, which means $\lambda \sim 10^{17}$, while $R \sim 10^{11}$ cm; therefore ordinary stars are transparent.

Although, in the late stages of evolution, the center of a large star consists of medium-weight elements, when stellar collapse occurs, a core composed predominantly of neutron matter begins to form at an approximate density of 10^{11} – 10^{12} gm cm⁻³ and radius $\sim 10^7$ cm. The mean free path is then $\lambda \approx 10^5$ – 10^6 cm, so that the system is opaque or optically thick. Neutrinos diffuse out of the core via multiple microscopic processes of production, absorption, and scattering. Because neutrinos remain the chief transporters of the $\sim 10^{53}$ ergs of energy liberated in the stellar collapse to form a neutron star, the neutrino-transport problem is critical for determining the mechanism of supernova explosions and whether the dense compact remnant is, in fact, a neutron star or a black hole.

The key problem here is not the onset of the collapse that follows from general

thermodynamic stability considerations and nuclear binding-energy systematics. The problem is to understand the reversal of the implosion after the central core has attained the mass ($1.4 M_{\odot}$) of the neutron-star pulsar remnant and the subsequent blowoff of the outer layers (containing most of the mass of the star), which constitutes the observed supernova explosion.

The weak-interaction neutral current affects stellar collapse simply because there are many more neutrino processes predicted than occur in the charged-current framework, and the implications of these new processes have received examinations (32–37). There are new processes, such as $\nu_e + n \rightarrow \nu_e + n$ and $\nu_e + p \rightarrow \nu_e + p$, which have cross sections in the collapsed core comparable to the traditional $\nu_e + n \rightarrow e^- + p$ and $\nu_e + e \rightarrow \nu_e + e$, and which do not suffer from the inhibition of the latter due to electron degeneracy. Similarly, ν_{μ} and $\bar{\nu}_{\mu}$ can now be produced efficiently in the hot core even at temperatures much lower than $kT \simeq m_{\mu}c^2$. For example, production occurs via the reaction $e^- + e^+ \rightarrow \nu_{\mu} + \bar{\nu}_{\mu}$, and these neutrinos can subsequently scatter via $\nu_{\mu} + e \rightarrow \nu_{\mu} + e$ and $\nu_{\mu} + n \rightarrow \nu_{\mu} + n$. These processes must be taken into account in treatments of neutrino transport.

An extremely simple process predicted by the weak neutral current, namely elastic scattering from a nucleus (Z, N) containing protons and N neutrons (33), can be important for neutrinos emerging from the core and traversing the mantle that contains nuclei such as Fe^{56} and Ni^{56} . The principle of quantum-mechanical coherence applies to this process and for a 20-MeV neutrino, a scattering cross section about a factor of $(Z + N)^2$ larger than that for single-nucleon processes is predicted. Because of this large cross section, momentum transfer from the neutrinos to the nuclei, i.e. neutrino-radiation pressure, may be quantitatively important in blowing off the outer layers of the star.

The purpose of this article is to review the properties of a weak neutral current with emphasis in the major processes of astrophysical interest, and to discuss our current understanding of its role in gravitational collapse and explosions. Both subjects are in a state of uncertainty. First, the detailed properties of the weak neutral current and therefore cross-section values are still not well determined experimentally. Second, the collapse of the hot stellar core and subsequent blowoff of the outer layers are intrinsically so complicated that it might be said that although Nature has clearly arranged for supernova explosions, man has not clearly understood the arrangements. Because of these uncertainties, this is not a detailed review whose main function is bibliographic. Instead, the main qualitative ideas are stressed at a level consistent with the interdisciplinary character of the subject. An astrophysicist should be able to read this article and appreciate the relative sizes of cross sections for various important neutrino processes and how to calculate some of them, while a particle physicist should be able to learn the basic issues in the present description of stellar collapse, as well as the relation of supernova explosions to such general questions as cosmic rays and the abundance of the elements.

1.1 *Neutrinos and the Latter Stages of Stellar Evolution*

As mentioned above, neutrinos play a role throughout stellar evolution. For most of the star's life, they serve as an energy sink, completely removing some fraction of

the energy from nuclear burning in the star's core. A well-known attempt at detecting these neutrinos from a standard main-sequence star—the sun—is being carried out by Davis and his collaborators using a tank of C_2Cl_4 in the Homestake gold mine (38). However, no definitive results have been obtained to date. These neutrinos from normal stellar evolution result from standard charged-current Fermi processes and thus are not of prime interest in this present review.

In the latter stages of a star's evolution, it is no longer just a small fraction of the nuclear energy that goes into neutrinos, but instead stellar energy loss becomes dominated by neutrinos rather than photons. Let us now briefly review the current ideas on stellar evolution [see also (39)], so that we can gain some perspective on the important role of neutrinos and see how they might even be the key to understanding how, at a massive star's death, it somehow manages to eject its outer material while leaving behind a dense neutron-star or black-hole remnant.

It is known that when stars first form from a gas of hydrogen and helium, the hydrogen burns in the core to helium ($4H \rightarrow {}^4He + 2e^+ + 2\nu_e + \text{energy}$) as the star evolves along the main sequence. When the helium core from such burning becomes large enough, it will begin to gravitationally contract until it heats to a sufficiently high temperature that the helium can begin to burn. During this core contraction, the outer envelope of the star expands and the star becomes a red giant. Then the core is supported by the thermal pressure at the temperature of helium burning, with the energy being supplied by the helium burning. When the helium burns, it creates carbon and oxygen via the triple- α reaction ($3 {}^4He \rightarrow {}^{12}C + \gamma$) and the ${}^{12}C(\alpha, \gamma) {}^{16}O$ reaction. Following the carbon core development, there is a branching point in the stellar evolution. If the carbon core is sufficiently massive, it too can reach a point where it will collapse, carbon will begin to burn and the star can proceed to the later stages of evolution, going successively to a neon core, an oxygen core, a silicon core, and finally to an iron-nickel core (40). However, if the core is not sufficiently massive, in particular, if it is of very low mass, the evolution could stop at this point, since sufficient pressure to support the entire star can come from degenerate electrons, in which case one would have a white dwarf (41). Thus, depending on the mass of the star, there is a branching point in the stellar evolution. For this review, it is important to note that for all the stages of evolution from carbon burning onward, the main energy-loss mechanism from the core is via neutrinos. This means that cores evolve on much more rapid time scales than if they had to wait for the photons to diffuse out. Thus, the cores evolve almost independently of the outer parts of the star.

STELLAR MASS RANGES Let us divide the possible mass ranges in which a star might fall into four segments. Stars with a mass less than M_{MS} require times longer than the present age of universe to evolve off the main sequence. Although these stars will eventually end their lives as white dwarfs, the time required for them to reach that stage is excessively long. The next division is the range between M_{MS} and some mass M_{WD} , that range where stars will be able eventually to become white dwarfs. Now, M_{WD} is not the Chandrasekhar mass, $1.4 M_\odot$, although $1.4 M_\odot$ is the upper limit for the mass of a white dwarf, since M_{WD} can be significantly greater than 1.4

due to mass-loss processes during the star's lifetime. Between M_{WD} and M_{CD} is a region where the carbon in the core will ignite under degenerate conditions; that is, the core is being partially supported by degenerate electron pressure so there is no feedback on the pressure when the carbon begins to burn. When this happens, one may get the carbon-detonation supernova model (e.g. 42). There is currently some question as to whether a detonation wave or a deflagration occurs and what the results of stars igniting carbon under these conditions will be (cf 43, 44). However, for completeness, this range must be mentioned. It may be that M_{WD} is greater than M_{CD} in which case this range would not exist at all. The third mass range, between M_{CD} and M_{pair} , is where the carbon ignites nondegenerately and goes on to burn to neon, the neon burns to oxygen, the oxygen burns to silicon, and the silicon burns to iron and nickel, so that one obtains the onion-skin model (see Figure 1). This is the mass range that is of primary interest in this review. As we show below, stars in this range may be able to explain the origin of the bulk of the heavy elements, cosmic rays, pulsars (neutron stars), and maybe even black holes. There is also the range beyond M_{pair} , where the pair-formation supernova occurs (45–48). However, M_{pair} is probably around $60\text{--}70 M_{\odot}$ and it is not at all clear that there are stars in this range, since none are seen. Of course, it may be that stars this massive always exist in clouds and thus it would be difficult to see them. In addition, it may be that such massive stars undergo such rapid mass loss that they never form a core large enough for the pair-formation supernova to take place (49). However, these very massive stars are believed to blow up via what is called the pair-supernova mechanism (45–47), assuming they would form and that they do not have as large a mass-loss rate as is implied by extrapolations up into that region. The range that

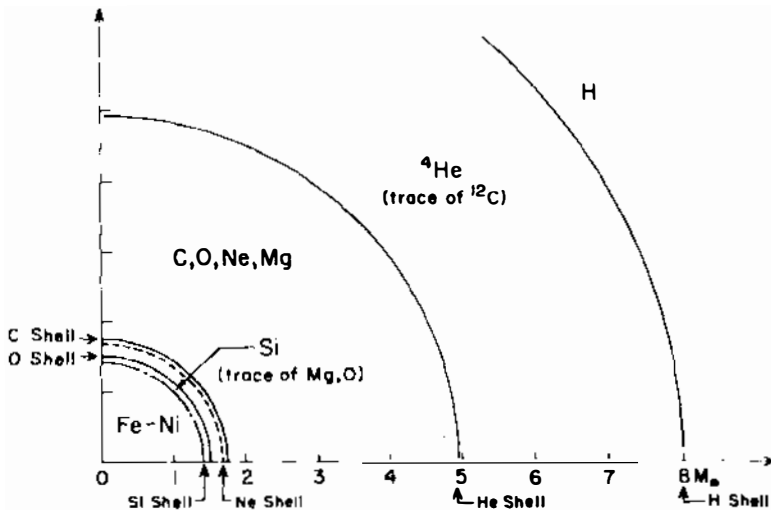


Figure 1 Cross section of an evolved $22\text{-}M_{\odot}$ star showing the $1.4\text{-}M_{\odot}$ Fe-Ni core with the surrounding shells of Si, O, Ne, C, He, and the H envelope.

is relevant for our current interest, though, is the range between M_{CD} and M_{pair} , where the star evolves into an onion-skin model with a very dense iron-nickel core, which then goes on to collapse gravitationally and produce many neutrinos.

Let us see what evidence there is concerning the characteristic masses M_{SD} , M_{WD} , M_{CD} , and M_{pair} . It is known that M_{MS} is somewhat less than $1 M_{\odot}$, since we know the main-sequence evolution times. M_{WD} is a mass for which a star is able to shed enough mass, during its lifetime to get below the $1.4 M_{\odot}$ Chandrasekhar limit. From the fact there is a white dwarf in the very young cluster, the Pleiades, and that all stars up to about $6 M_{\odot}$ are still on the main sequence in the Pleiades, it seems clear that for that particular white dwarf, a star greater than $6 M_{\odot}$ was able to shed enough mass to become a white dwarf. However, that does not mean that all stars from $6 M_{\odot}$ down shed enough mass to become white dwarfs. There may be other variables that enter in, such as rotation and magnetic fields, which alter the rate of mass loss for different stars. For example, one finds that the number of white dwarfs in the Hyades is consistent more with a value for M_{WD} of around $3-4 M_{\odot}$, than a value of $6 M_{\odot}$. However, there may have been other factors contributing. For example, white dwarfs may have been lost from the Hyades, in which case the mass limit would go above $4 M_{\odot}$. In view of the present uncertainties, it is probably best to estimate M_{WD} as somewhere between 4 and $8 M_{\odot}$, with 6 being a reasonable best guess. The next mass limit is where carbon ignites under non-degenerate conditions, as contrasted with lower-mass cores having carbon that ignites under degenerate conditions. If we take the current estimates for the rate of the $^{12}\text{C} + ^{12}\text{C}$ reaction and the estimates of Paczynski (50) on the mass of the carbon core associated with the main-sequence mass, then we estimate the mass at which carbon burns nondegenerately to be approximately $8 M_{\odot}$. However, because the mass of the carbon core associated with the main-sequence mass depends on convection theory, this mass is very uncertain. In fact, recent calculations of Barkat (48) show that this mass may go down to 5 or $6 M_{\odot}$. Also, the rate of the $^{12}\text{C} + ^{12}\text{C}$ reaction is somewhat uncertain [cf the discussion in (39)], which also affects the location of this mass limit. Therefore it is best to just say that M_{CD} is somewhere between $4 M_{\odot}$ and $10 M_{\odot}$, with a "best" value of about 8 . Note that there is an overlap between M_{WD} and M_{CD} . That is, there may be no stars in the range that undergo carbon detonation. In fact, it may very well be that the phenomenon of a degenerate core does contribute to the mass-loss phenomenon, in which case M_{WD} and M_{CD} may be the same number. This argument has been discussed by Paczynski, among others. It is actually very good that this mass region for the carbon-detonation model is small or nonexistent, in that current calculations of the carbon detonation, if the carbon really does detonate, indicate that the entire carbon core is converted to iron and ejected. Were ejection to happen for a very large-mass region, then there would be too much iron in the galaxy and we would have an iron catastrophe. Therefore, it is important either that this mass region be very small or that the final stage of life of these stars not be a carbon-detonation supernova.

The upper mass limit on the onion-skin model occurs when the carbon-oxygen core becomes sufficiently large that rapid pair formation causes a dynamic instability. Dearborn (49) has recently shown that any reasonable extrapolation

Table 1 Stellar mass ranges

Mass range	Consequence
$M < M_{\text{MS}} \lesssim 1 M_{\odot}$	Main sequence life $>$ age of universe
$M_{\text{MS}} \lesssim M \lesssim M_{\text{WD}} \approx 6 \pm 2 M_{\odot}$	Star loses sufficient mass to become a white dwarf of less than $1.4 M_{\odot}$
$M_{\text{WD}} \lesssim M \lesssim M_{\text{CD}} \approx 8 \pm \frac{4}{3} M_{\odot}$	Carbon ignites under degenerate core conditions → Carbon-detonation supernova?
$M_{\text{CD}} \lesssim M \lesssim M_{\text{pair}} \approx 100 \pm 30 M_{\odot}$	Onion-skin star develops finally, yielding Fe core surrounded by Si, O, Ne, C, He, and H shells → Core-collapse supernova
$M \gtrsim M_{\text{pair}}$	May not occur, due to such high mass loss that core never deviates from onion-skin scenario → Pair-formation supernova

of main-sequence mass-loss rates into the regime of massive stars would indicate that these massive stars lose mass at such a rapid rate that cores sufficiently massive for the pair-formation scenario may never develop. However, it is difficult to be certain of properties of these unseen massive stars. Thus, to be conservative, we put M_{pair} at $\sim 100 M_{\odot}$. Table 1 summarizes the above comments [see also the review by Tinsley (51)].

SUPERNOVAE There is common feeling in astrophysics that the observed supernovae may be related in name only to the theoretical supernovae. This misconception occurs to a large degree because, until recently, theoreticians had not tried to make detailed connections between explosion mechanisms that originate in the interior of the star and observational characteristics of supernovae that are determined by the expansion of the hot surface material.

Recently, this has changed, as several groups (52–54) have begun to calculate what happens when a great deal of energy is released in the center of a big gas cloud. A supernova is in fact just that, since it has a very-large-radius, low-density envelope with a very small, compact stellar core from which the energy comes. The calculations of Lasher (53), Chevalier (54), and Falk & Arnett (52) seem to be able to fit many of the observed characteristics of supernovae. In particular, they find that less than 1% of the neutron-star binding energy is needed to produce an observed supernova. However, one thing they are not able to resolve is how this energy is to be deposited, since by the time the supernova light is seen, almost all memory of how the “bomb” was set off in the middle of the cloud disappears. (Perhaps neutrino astronomy will one day resolve this problem.)

In the rest of this review, we focus on the details of this energy release. First, we

recall some of the observed properties of supernovae. One of the most interesting is the rate of appearance of supernovae. Tammann (55) has done a systematic study and points out that, contrary to folklore, the supernova rate in our galaxy seems to be about one every $15 \pm \frac{1}{2}^5$ years. This high rate comes from the fact that we can only see supernovae in a narrow segment of our galaxy, so the fact that only 5 have been observed in the past 1000 years should not be discouraging. Tammann goes on to show that this is consistent with observed supernovae rates in other galaxies similar to our own. It has also been noted (56) that pulsar-formation rates may be much higher than previously thought. Since pulsars are presumably neutron stars left by supernovae, then pulsar-formation rates should be very closely related to supernovae rates.

Another observational point that tends to confuse physicists is the term Type I or Type II supernovae, which refers to the observational characteristics of the light curve, i.e. light intensity as a function of time after the explosion. Type I supernovae have light curves of a special shape with a peak and then an exponential drop [cf (57)], and Type II supernovae are a more loosely defined grouping of everything else (50). [Zwicky (58) originally employed many more supernovae classifications, but Type I and Type II are the only ones in frequent usage.] Type I's also show very little or no hydrogen, whereas Type II's show high levels of hydrogen. Recently Tammann (55) has argued that the light curves and energetic characteristics of one type seem to blend into those of the other type. Thus, the only clear distinction may be one of the spectrum. The location of young stars near the inner edge of galactic spiral arms suggests (51) that Type II supernovae probably have massive stars as their progenitors. The situation for Type I's is not so clear. At one time it was assumed that since Type I's occur in elliptical galaxies that normally are thought to have old stellar populations, then Type I's must be low-mass objects. However, objections have been raised to that argument (51, 55). It may be that a Type I results whenever the hydrogen envelope is stripped from the core. This may occur in the course of binary star evolution or by some other means. It may also be that Type I's are related to carbon-detonation models and Type II's to more massive stars. (Carbon detonation might occur by accretion in binary evolution, even if it is bypassed by mass loss in single-star evolution.)

In 1054 A.D., Chinese observers recorded the appearance of the most famous supernova, which was bright enough to be seen during the day. It is not easy to tell from the remnant we observe today, the Crab Nebula, whether the supernova was Type I or Type II. Of particular importance is the fact that the Crab's spectrum is dominated by helium. Arnett (59) has used this point to argue that the progenitor for the Crab was a $10\text{--}12 M_{\odot}$ star, since such stars should produce large amounts of helium. The mass of the Crab remnant is very uncertain (60). It is greater than a solar mass, and could be as large as 8 or $10 M_{\odot}$. It is also conceivable that the progenitor of the Crab lost large amounts of mass (its hydrogen) prior to the explosion. The most exciting aspect of the Crab is the existence of the pulsar at its center. This observation is the prime reason why one wants a supernova mechanism that not only blows off the outer part of a star, but also allows the central region to collapse into a neutron star.

Another interesting supernova type is the “invisible supernova,” of which the remnant Cas A seems to be an example (61). It went off ~ 300 years ago, but its appearance was never recorded on earth. This is all the more interesting when one notes that Cas A shows evidence for nucleosynthesis of heavy elements, and Chevalier argues that it is a good cosmic-ray source. Arnett (59) has argued that the progenitor for Cas A was a massive star of $\sim 24 M_{\odot}$. Perhaps some mass-loss mechanism so stripped Cas A that it had no envelope to produce a bright optical light curve when it went off.

Supernovae observations are rapidly improving and more calculations are being done to relate core explosions to exterior light. Thus, the relation between observed and theoretical supernovae seems to be improving.

NUCLEOSYNTHESIS The stars in which we are most interested are those that do evolve to an onion-skin model with an iron-nickel core surrounded by a silicon shell, an oxygen shell, neon shell, carbon shell, helium shell, and a hydrogen envelope, and that may become Type II supernovae. One particularly attractive feature about these more massive stars is that they produce supernovae and they seem to be able to provide us with the synthesis sites for the bulk of the heavy elements (40, 62). In fact, they also seem to be able to provide good sources for the galactic cosmic rays (62, 63). In particular, with regard to the production of the heavy elements, it is interesting that the conditions in these massive stars may very well be close to those conditions achieved in the explosive nucleosynthesis calculations.

In the late 1960s and early 1970s, much work on explosive nucleosynthesis was carried out. This work has been reviewed recently by Arnett (64), and discussed in the book *Explosive Nucleosynthesis* (42), and so is not described in detail here. What was done in these explosive nucleosynthesis calculations was to set up a nuclear-reaction network that coupled all of the reactions thought to be important in a particular region of the chart of the nuclides. The reaction rates were dependent on temperature and density. The temperature and density in these calculations were made time-varying, following a simple exponential expansion law $\rho = \rho_0 \exp(-t/\tau)$, where the expansion rate was a multiple of the free-fall time $1/(24\pi G\rho)^{1/2}$. These calculations also tended to assume that the temperature varied in an adiabatic manner, with ρ proportional to T^3 . It was found that good agreement with the observed abundances was obtained only for a certain set of conditions—values of T_0 and ρ_0 and the composition variable η_0 that was just a measure of the neutron enrichment [η is equal to $(n-p)/(n+p)$]. However, if ρ , T , and η were away from these values, then agreement was not possible. In particular, Table 2A shows the temperature and density fits that were obtained for the conditions of explosive carbon burning, explosive oxygen burning, and the equilibrium process.

The fits for these three processes enable one to explain the origin of all the elements from neon through the iron peak if these processes really occurred in a supernova. It was found that in the standard course of stellar evolution, the main α -particle nuclei were produced in roughly the right abundances in the onion-skin models (40, 62). However, the non- α -particle nuclei were not produced in the main-line course of stellar evolution. Table 2B shows the temperatures and densities associated

Table 2a Explosive nucleosynthesis conditions

Process	Explosive C burning	Explosive O burning	e-Process
Fits	Ne-Si	Si-Ca	Fe peak
$T_0(\text{K})$	2×10^9	$3.1\text{--}3.9 \times 10^9$	$\geq 10^9$
$\rho_0(\text{g cm}^{-3})$	10^{4-7}	10^{5-7}	$\geq 5 \times 10^7$
η	0.001–0.004	0.002–0.004	0.002–0.004

2b Mantle of $22\text{-}M_\odot$ star ($8\text{-}M_\odot$ evolved He core)

Zone	C	Ne	O	Si
T	1.1×10^9	1.1×10^9	2.8×10^9	4.0×10^9
" T "	1.8×10^9	1.8×10^9	4.0×10^9	7.0×10^9
ρ	9.4×10^9	1.1×10^5	2.8×10^6	5.9×10^9
η	≥ 0.0011	0.0019–0.0027	0.0019–0.0027	$> 0.0019\text{--}0.0027$

with the different zones in a $22\text{-}M_\odot$ star. Notice that these temperatures are not the same as those needed in the explosive burning conditions to fit the non- α -particle nuclei. However, the densities are roughly right. It is interesting to note that if a shock were propagated through the star to turn it into a supernova, then the temperatures in those zones would be changed. In fact, if we make the shock necessary to turn one of these massive stars into an observed Type II supernova, then we can calculate what the effective temperature would be in each of these zones, and we find that the temperatures obtained, labeled " T " in Table 2B, are very close to those temperatures that one estimates from the explosive nucleosynthesis calculations. Thus it seems that if these massive stars do make the observed supernovae, then the ejecta from these stars will produce the elements from carbon to iron in their correct abundances, not just the α -particle nuclei, but all the nuclei, since the conditions would be those of explosive nucleosynthesis.

The above argument used a $22\text{-}M_\odot$ star; however, it should be remembered that each different-mass star produces a slightly different composition. Figure 2 shows the density profiles of a $22\text{-}M_\odot$ and a $15\text{-}M_\odot$ star. Notice that the central core regions of both of these stars have almost identical iron-nickel cores. This core convergence results from the fact that neutrinos are the dominant energy-loss mechanisms for the cores, which thus evolve independently of the rest of the star at each neutrino-dominated stage. This tends to mean a $1.4\text{-}M_\odot$ core is formed. When additional material is added to the core, it is pushed over the Chandrasekhar limit and collapse occurs. Since each composition zone in the star is roughly at the same density regardless of the mass of the star, the more massive star will have much more carbon and oxygen than the lower-mass star, whereas the lower-mass star will tend to have much more helium relative to the other material in the star. Therefore, if one blows up the $15\text{-}M_\odot$ star, one would get a different ejecta than if one blows up the $22\text{-}M_\odot$ star. We have already shown that the $22\text{-}M_\odot$ star, if it becomes an observable

supernova, will yield ejecta that produce an abundance of elements that is very similar to the observed abundance in the solar system. In fact, from the Dearborn (49) mass-loss estimates, it can be argued that most stars more massive than this will end up with a very similar internal structure due to their large main-sequence mass loss.

If one does not weight by the amount of mass ejected, then the mass of the typical star that blows up is $12\text{--}15 M_{\odot}$. Thus a typical supernova would not produce as many heavy elements, but would eject relatively more helium than the $22\text{-}M_{\odot}$ star. If you look at the Crab Nebula, it has a large amount of helium; thus it may be an example of a lower-mass star rather than of the more massive star.

It has also been demonstrated (62, 63) that the cosmic rays may be more like the typical supernova than like the mass-weighted supernova, and some of the differences in composition between the solar-system abundances and the cosmic rays may be due to just this different way in which the cosmic rays weight their selection of supernovae.

CORE EVOLUTION We have seen that all these massive stars have a very similar core structure; thus, if we can understand what happens in the core, we might be able to understand how these stars blow up. Let us look at the evolution of the core once it reaches the point of being an iron-nickel core of about $1.4 M_{\odot}$. All of the cores tend to pass through a point with the temperature around 10^{10} °K and the density around

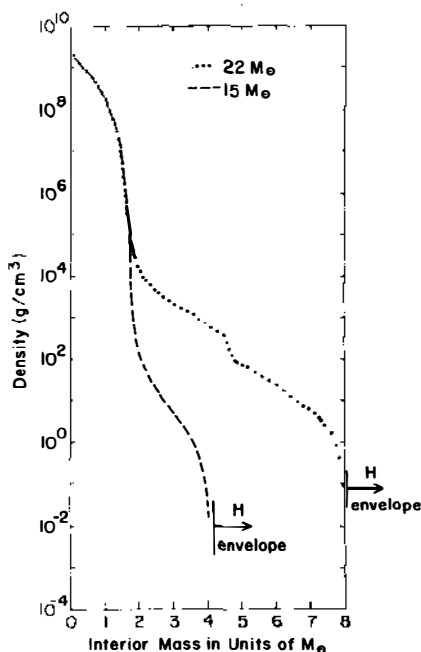


Figure 2 The density profile of a $\sim 22\text{-}M_{\odot}$ and a $\sim 15\text{-}M_{\odot}$ star.

$10^{10} \text{ g cm}^{-3}$. They are supported by degenerate electrons with an adiabatic index $\gamma = 4/3$. Around this point, four things begin to happen:

1. Electron captures occur, electrons being captured due to their high chemical potential. Each electron capture causes a neutrino to be emitted; capturing electrons remove the electrons that are supporting the core.
2. Photodisintegration occurs, breaking the iron and nickel down into alphas, and then eventually down into free neutrons and protons.
3. The surrounding silicon shell continues to burn and thus adds more iron and nickel to the core, increasing the core mass to greater than $1.4 M_{\odot}$.
4. The core is extremely hot, so neutrino pair emission takes place, carrying more energy out of the core.

All four of these processes yield the net effect of the core collapsing. Eventually the collapsing core will switch over from its adiabatic index γ being dominated by the electron-degeneracy pressure to the pressure being dominated by a thermal neutron gas with a γ of $5/3$. This transition from a γ of $4/3$ to a stiffer γ of $5/3$ will cause a hydrodynamic bounce (65). It is interesting to note that the mass of the core inside of the bounce region always seems to be greater than about $1.1 M_{\odot}$. This implies that one can never eject that inner $1.1 M_{\odot}$, which would always go to form a remnant. Thus it seems that the mass of any remnant must be greater than $1.1 M_{\odot}$. It may be significantly greater and go on to become a black hole, but in any event there should be no neutron stars with masses less than $1.1 M_{\odot}$ as a result of the core collapse of these massive stars. This is in good agreement with the observations of the masses of neutron stars (66).

What happens next in the collapse is a matter about which there is currently much uncertainty. It is clear that the manner in which the core releases the bulk of its binding energy is via neutrinos. Thus, in order to form a neutron star, the star must release $\sim 10^{53}$ ergs of neutrinos. Whether or not these neutrinos are what cause the explosion, the ejection of the outer part of the star is a matter of some debate. However, it is important to realize that the neutrinos invariably seem to carry the bulk of the binding energy regardless of whether or not they cause the explosion. It seems clear that if one wants to observe gravitational collapse to the formation of neutron stars and black holes, one should look where the bulk of the energy is, in the neutrinos.

Rather than go into the complete collapse scenario now, let us just stop and examine the central-core configuration as this final gravitational collapse occurs. What we have is a hot, dense central region of neutron gas surrounded by a shell of alphas and neutrons, which in turn is surrounded by a shell of iron and nickel. This comprises the $\sim 1.4 M_{\odot}$ core. The surrounding Si, O, Ne, C, He, and H shells are at progressively lower densities and are not considered part of the core. One great possibility that weak neutral currents introduce is coherent neutrino scattering with a cross section approximately proportional to the square of the atomic weight (33). It is apparent that such a dependence might help in getting high neutrino fluxes through the $A = 1$ neutronized region to deposit momentum into the $A \sim 60$ iron-nickel shell. In principle, this could be the mechanism that enables the central core to collapse and form a neutron star while the outer material, containing the newly synthesized elements from carbon to iron, is ejected.

In the remainder of this review, we discuss the microscopic weak-interaction physics that is so important in understanding the collapse, and may even be important in understanding the ejection of the outer material. Following this discussion of weak-interaction physics, we discuss the gravitational-collapse scenario of the core.

The study of core collapse is an extremely active field at the present time. Thus the point of this review is not to emphasize any particular set of detailed core-collapse calculations that may be supplanted tomorrow, but rather to provide an overview of the important weak-interaction physics, particularly that concerning the neutral current.

1.2 Coherent Nuclear Scattering

Situations in which simple quantum-mechanical principles have macroscopic effects have a special appeal because they illustrate the unity of science. Electron-degeneracy pressure, which led to Chandrasekhar's prediction of the properties of white-dwarf stars, is one well-established astrophysical example. Stellar collapse may contain another example if, as current ideas indicate, nuclear elastic scattering $\nu + (Z, N) \rightarrow \nu + (Z, N)$ does indeed play a special role. The enhanced cross section for this process is due to the basic principle of quantum-mechanical coherence; it is worth discussing the coherent-scattering phenomenon to illustrate its generality.

In its simplest form, the phenomenon occurs whenever an elementary projectile (in our case the neutrino) scatters elastically from a composite system (in our case the nucleus), assumed to consist of A individual constituents (the nucleons) at positions $\mathbf{x}_i$, $i = 1, 2, \dots, A$. Due to the superposition principle, the amplitude $F(\mathbf{k}', \mathbf{k})$ for scattering from an incident momentum $\mathbf{k}$ to an outgoing momentum $\mathbf{k}'$ can be written as the sum of the sum of contributions from each constituent:

$$F(\mathbf{k}', \mathbf{k}) = \sum_{j=1}^A f_j(\mathbf{k}', \mathbf{k}) \exp i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{x}_j, \quad 1.1$$

where the individual amplitudes $f_j(\mathbf{k}', \mathbf{k})$ are added with a phase factor that takes into account the relative phase of the wave scattering at $\mathbf{x}_j$. This expression is valid under the simplifying assumption that individual amplitudes are small enough to neglect multiple scattering. This is extremely well satisfied for neutrinos!

The differential cross section is

$$\frac{d\sigma}{d\Omega} = \left| F(\mathbf{k}', \mathbf{k}) \right|^2, \quad 1.2$$

and its size depends on the relation between the momentum transfer $\mathbf{q} = \mathbf{k}' - \mathbf{k}$, ($q = |\mathbf{q}| = 2k \sin \theta/2$, where k is the beam momentum and θ the scattering angle) and the size R of the composite system ($R = \max_{i,j} |\mathbf{x}_i - \mathbf{x}_j|$).

If the dimensionless quantity qR is of order 1 or greater, then the relative phase factors are important, and contributions from individual scatterers tend to cancel (in the absence of special spatial relations among the position vectors $\mathbf{x}_i$ such as occur in a crystal). The elastic scattering cross section is therefore expected to be small.

For momentum transfer small compared to inverse target size, i.e. $qR \ll 1$, the

relative phase factors are inconsequential and contributions from individual scatters add coherently, yielding

$$\frac{d\sigma}{d\Omega} \simeq A^2 |\bar{f}(\mathbf{k}', \mathbf{k})|^2, \quad 1.3$$

where $\bar{f}(\mathbf{k}', \mathbf{k})$ is an amplitude averaged over all constituents,

$$\bar{f}(\mathbf{k}', \mathbf{k}) = \frac{1}{A} \sum_j f_j(\mathbf{k}', \mathbf{k}). \quad 1.4$$

If there is only one type of constituent, then $\bar{f}(\mathbf{k}', \mathbf{k}) = f_j(\mathbf{k}', \mathbf{k})$, and the coherent scattering cross section is A^2 times the cross section for scattering from a single constituent.

If individual constituent scattering amplitudes differ, as is generally expected for the protons and neutrons of different spin projections in a nucleus, there can be cancellations among the contributions to Equation 1.4. No rigorous statement, therefore, can be made about the size of the coherent cross section without knowledge of the detailed scattering amplitudes involved. However, barring extreme cancellation, it is expected that the coherent cross section is approximately A^2 times a typical single constituent cross section.

To clarify the physical ideas involved, we assumed in the previous discussion that individual constituents were located at definite points $\mathbf{X}_j$ in the composite target. This restriction is inessential and the discussion can be generalized to the case where the target is treated as a quantum-mechanical system. There is one point of principle concerning the various quantum states, such as charge and spin projection, of the individual constituents of the composite system: namely, the instruction to sum over constituent amplitudes in Equation 1.1 follows from the superposition principle only if there is no change in quantum states. Equation 1.1 does not apply to spin-flip and charge-exchange processes, and these cross sections are not amplified by a factor of A^2 in a composite target.

Further insight may be gained by introducing the spatial density of scatterers $\rho(\mathbf{x}) = \sum_j \delta(\mathbf{x} - \mathbf{x}_j)$. In the case of one type of constituent, we can drop the subscript j and Equation 1.1 becomes

$$F(\mathbf{k}', \mathbf{k}) = f(\mathbf{k}', \mathbf{k}) \int d^3 \times \exp(i\mathbf{q} \cdot \mathbf{x}) \rho(\mathbf{x}). \quad 1.5$$

Coherent scattering therefore measures the form factor of the composite target, i.e. the Fourier transform of its density. This interpretation is very general and applicable to neutrino-nucleus coherent scattering, which we calculate in Section 2.4.

The spherically symmetric Gaussian density $\rho(r) = A(4\pi R^2)^{-3/2} \exp(-\frac{1}{4}R^{-2}r^2)$, with $R = r_0 A^{1/3}$ and $r_0 \simeq 10^{-13}$ cm illustrates the systematics of nuclei of $A = Z + N$ constituents as mass number varies. The form factor corresponding to this density is $\rho(q) = A \exp(-q^2 R^2)$. We now discuss the qualitative behavior of coherent scattering cross sections for such nuclei.

For fixed beam momentum k , the maximum momentum transfer is $q_{\max} = 2k$. For moderate or large energies ($2kR \gtrsim 1$), one finds that $d\sigma/d\Omega$ has a forward peak

and falls off sharply at large angles or momentum transfer because of the form factor. Under these conditions the integrated elastic cross section

$$\sigma_{el} = \int d\Omega \frac{d\sigma}{d\Omega} \quad 1.6$$

has the undramatic A dependence $A^{4/3}$.

However, at low energy, $2kR \ll 1$, the condition of coherence is satisfied for all angles, and the angular dependence is determined by the constituent amplitude $f(\mathbf{k}', \mathbf{k})$ in Equation 1.5 because the form factor is very nearly constant. The full benefits of coherence are therefore realized at low energies and $\sigma_{el} \propto A^2$.

These conclusions are largely independent of the nature of the projectile, whether pion, electron, or neutrino. In Fe^{56} , the radius is $R^{-1} \simeq 80 \text{ MeV}$ ($c = \hbar = 1$), so the condition for full coherence in the scattering of neutrinos in the mantle surrounding a collapsed stellar core is $E_\nu \ll 40 \text{ MeV}$, which is reasonably well satisfied for the 20-MeV neutrinos expected in stellar collapse.

Finally, we note that removal of some of the simplifying assumptions of our discussion actually leads to broader application for the coherent scattering concept. When individual scattering amplitudes are large, terms describing multiple scattering in the target must be added in Equation 1.1. The multiple-scattering formalism describes π -nucleus scattering very well and has been reviewed recently (67, 68). In crystals and large molecules, maxima can occur in Equation 1.1 at nonzero momentum transfer because of the regular spacing of constituents. This is just the Bragg scattering phenomenon that has been fruitfully applied to determine the structure of biological molecules by X -ray scattering. Electron scattering in nuclei [for a recent review see (69)] is a very-well-studied application of the coherent scattering formalism. It is similar to the neutrino-scattering case, both because the single-scattering approximation is usually valid and because both electrons and neutrinos couple to nucleons through vector (and axial vector) currents.

2 WEAK INTERACTIONS

2.1 Charged-Current Processes

The formalism conventionally used to describe charged-current weak interactions involves a phenomenological or effective Lagrangian that is a product of local current operators. Amplitudes for weak decays and scattering processes are calculated in the lowest order of perturbation theory and are in good agreement with experimental data. However, the theory cannot be fundamental because it is non-renormalizable, and higher-order corrections to physical amplitudes are infinite. An alternate hypothesis is that the charged-current weak interactions are mediated by a charged intermediate vector boson of mass m_w . For momentum transfers small compared to m_w , the experimental predictions are the same as for the current-current theory. The intermediate vector-boson theory is also non-renormalizable.

Renormalizability problems of the weak interactions are solved in the non-Abelian gauge theories (26–29), such as those first proposed by Weinberg (20) and Salam (21) and many more recent models. In these theories there are one or more massive

charged vector bosons (W 's) that mediate charged-current weak interactions. The electromagnetic interactions mediated by the massless photon A_μ are naturally included, and there are generally one or more massive neutral-vector particles (Z 's) that mediate neutral-current processes. The W and Z bosons are predicted to have masses on the order of 50 GeV or more, which is consistent with experimental evidence that suggests that the W boson, if it exists, has a mass greater than 20 GeV. In addition to vector particles, there are spinless Higgs mesons that are important for the mathematical structure of gauge theories, but have more subtle experimental effects.

The fundamental fermion fields of the gauge theories are those that correspond to leptons such as e , ν_e , μ , and ν_μ , and quarks, which are presumed to be the constituents of hadrons such as p and n . Currents J_λ^W and J_λ^Z , which are bilinear in the fermion fields, determine the observed properties of weak processes. The properties of processes of astrophysical interest and even most processes measured at present accelerators are insensitive to the full details of the gauge theories and may be described by current-current effective Lagrangians, which we use in this paper.

For charged-current processes, the effective Lagrangian is

$$\mathcal{L} = (2)^{-1/2} G J_\lambda^W (J_\lambda^W)^\dagger, \quad 2.1$$

$$G = (1.015 \pm 0.03) \times 10^{-5} m_p^{-2},$$

and the current is decomposed into leptonic and hadronic parts:

$$J_\lambda^W = l_\lambda^W + h_\lambda^W, \quad 2.2$$

with

$$l_\lambda^W = e_e \gamma_\lambda (1 - \gamma_5) \nu_e + \bar{\mu} \gamma_\lambda (1 - \gamma_5) \nu_\mu, \quad 2.3$$

where we use the γ matrices

$$\gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \gamma = \begin{pmatrix} 0 & \sigma \\ -\sigma & 0 \end{pmatrix}, \quad \gamma_5 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

and σ_1 , σ_2 , and σ_3 are the 2×2 Pauli matrices. Because of the extremely low experimental limits on the neutrino mass,

$$m_{\nu_i} < 60 \text{ eV} \quad m_{\nu_\mu} < 0.65 \text{ MeV},$$

the neutrinos are assumed to be massless. The matrix $(1 - \gamma_5)$ projects out the left-handed chirality components of fermion fields. For the massless neutrino, this has the implication that the neutrinos that participate in charged-current processes are purely left-handed, while antineutrinos are right-handed.

A particle physicist would write the hadronic current in terms of quark fields, and there is strong theoretical and experimental support for four quarks u , d , s , and c , whose quantum numbers are listed in Table 3. The weak hadronic charged current is then

$$h_\lambda^W = d_c \gamma_\lambda (1 - \gamma_5) u + \lambda^c \gamma_\lambda (1 - \gamma_5) c. \quad 2.4$$

Table 3 Quantum numbers of the four quarks: u , d , s , and c

Quarks	Electric charge (Q)	Baryon number (B)	Strangeness (S)	Charm (C)
u (up)	$+\frac{2}{3}$	$\frac{1}{3}$	0	0
d (down)	$-\frac{1}{3}$	$\frac{1}{3}$	0	0
s (strange)	$-\frac{1}{3}$	$\frac{1}{3}$	-1	0
c (charm)	$+\frac{2}{3}$	$\frac{1}{3}$	0	+1

There is a striking analogy with the form of l_λ^W , in accord with the idea that all fundamental fermion fields participate in the weak interaction with universal strength. Universality is in agreement with experiment if the fields d_c and λ_c are rotated by an angle θ_c with respect to d and λ :

$$d_c = \cos \theta_c d + \sin \theta_c \lambda,$$

$$\lambda_c = \sin \theta_c d + \cos \theta_c \lambda,$$

which is the quark-model form of a formulation of universality due to Cabibbo (70). Experiment gives the result $\cos \theta = 0.976$, $\sin \theta = 0.218$.

Ordinary β decay and the related reactions

$$n \rightarrow p + e^- + \bar{\nu}_e,$$

$$e^- + p \rightleftharpoons \nu_e + n, \quad 2.5$$

$$e^+ + n \rightleftharpoons \bar{\nu}_e + p$$

are important for stellar collapse. To describe them, it is sufficient to use a phenomenological term in the current h_λ^W of the form (26)

$$\gamma_\mu (g_V - g_A \gamma_5) p, \quad 2.6$$

with $g_V = 1$, $g_A \approx 1.21$. This form is consistent with Equation 2.4 and was suggested by the conserved vector-current (CVC) hypothesis (13–15). According to the CVC idea, the strangeness-conserving ($\Delta S = 0$) part of h_λ^W is a component of the isotopic spin current.

In this review we are only concerned with the $\Delta S = 0$ part of the hadronic current, parameterized as in Equation 2.6. Other $\Delta S = 1$ and $\Delta C = 1$ processes predicted by Equation 2.4 are irrelevant in stellar collapse because the temperatures ($kT < 30$ MeV) and densities ($\rho < 10^{14}$ gm cm $^{-3}$) are insufficient to produce strange and charmed particles. At the higher densities encountered in the collapsed state of the neutron star, the nucleon Fermi energies are high enough so that strange hyperons appear and are stable. However, the weak interaction is less important in determining the structure of the star at this point.

Reactions that play an important role in stellar collapse, for both charged and neutral currents, are listed in Table 4. Feynman diagrams for the processes in Equation 2.5 are given in Figure 3 as they would occur in a theory with intermediate

Table 4 Astrophysical weak processes

$e^- + p \rightleftharpoons \nu_e + n$	
$e^+ + n \rightleftharpoons \bar{\nu}_e + p$	unchanged by neutral currents
$e^- + \nu \rightarrow e^- + \nu$	
$e^+ + e^- \rightarrow \nu + \bar{\nu}$	
$\Gamma \rightarrow \nu + \bar{\nu}$	altered by neutral currents
$e^\pm + \gamma \rightarrow e^\pm + \nu + \bar{\nu}$	
$e^- + Z \rightarrow e^- + Z + \nu + \bar{\nu}$	
$n + n \rightarrow n + n + \nu + \bar{\nu}$	
$A^* \rightarrow A + \nu + \bar{\nu}$	
$\nu + \nu \rightarrow \nu + \nu$	
$\nu + p \rightarrow \nu + p$	allowed only by neutral currents
$\nu + n \rightarrow \nu + n$	
$\nu + A \rightarrow \nu + A$	

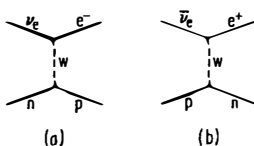
vector-boson exchange. At low energy, the amplitudes take the form suggested by the local-current effective Lagrangian.

2.2 Neutral-Current Processes

Neutral current processes can be divided into $\Delta S = 0$ and $\Delta S = 1$ reactions and decays. Prior to the 1973–74 experiments, upper limits on the $\Delta S = 0$ reactions were not very stringent, and experiments (1–7, 17–19) now suggest a strength of about 20% of corresponding charged-current reactions. The absence of weak decays such as $\Sigma^+ \rightarrow p + e^+ + e^-$, $K^+ \rightarrow \pi^+ + \nu + \bar{\nu}$ and $K^0 \rightarrow \mu^+ + \mu^-$ has long indicated that $\Delta S = 1$ neutral-current processes are strongly suppressed.

The Weinberg-Salam model was originally proposed as a theory of purely leptonic processes, but extensions to include hadrons have been given (71, 72). Both $\Delta S = 0$ and $\Delta S = 1$ neutral-current processes would occur in models based on the three quarks u, d, s . The $\Delta S = 1$ neutral current is eliminated very naturally if a fourth charmed quark c is added, as was suggested by Glashow, Maiani & Iliopoulos (73). The recent experimental discovery of charmed particles is a strong confirmation of this idea (74).

The Weinberg-Salam model is based on the gauge group $SU(2) \times U(1)$. It contains an arbitrary parameter $\tan \theta_w$, which is the ratio of the gauge coupling constant of the $U(1)$ and $SU(2)$ factors. The masses of the vector bosons W and Z which mediate

**Figure 3** Feynman diagrams for neutrino absorption.

charged- and neutral-current processes are, respectively,

$$m_W^2 = \frac{e^2(2)^{1/2}}{8G \sin^2 \theta_W},$$

$$m_Z^2 = \frac{e^2(2)^{1/2}}{2G \sin^2 2\theta_W},$$

where $e^2/4\pi = \frac{1}{137}$ is the fine-structure constant.

The form of the effective Lagrangian for neutral-current processes is not very well determined experimentally. For this reason, we present first a reasonably general expression in terms of vector and axial vector currents that is consistent with most gauge-theory models, and we then give the special choice of parameters corresponding to the Weinberg-Salam model. It is worth noting that even the V-A structure is not fully confirmed experimentally, and that theories with scalar, pseudoscalar, and tensor currents (75, 76) are difficult to rule out. Some astrophysical implications of S , P , and T neutral currents are discussed later.

The effective Lagrangian is

$$\begin{aligned} \mathcal{L} &= (2)^{-1/2} G J_\lambda^Z J^{Z\lambda}, \\ J_\lambda^Z &= l_\lambda^Z + h_\lambda^Z. \end{aligned} \quad 2.7$$

The leptonic current may be parameterized as

$$\begin{aligned} l_\lambda^Z &= \bar{\nu}_e \gamma_\lambda \frac{1}{2}(1 - \gamma_5) \nu_e + \bar{e} \gamma_\lambda (a_e + b_e \gamma_5) e \\ &\quad + c_\mu \bar{\nu}_\mu \frac{1}{2}(1 - \gamma_5) \nu_\mu + \bar{\mu} \gamma_\lambda (a_\mu + b_\mu \gamma_5) \mu \\ &\quad + \sum_L c_L \bar{\nu}_L \gamma_\mu (1 \pm \gamma_5) \nu_L + \bar{L} \gamma_\lambda (a_L + b_L \gamma_5) L. \end{aligned} \quad 2.8$$

Note that $\mu - e$ universality is not assumed. There is experimental evidence (108) for a charged heavy lepton and a possible associated neutrino. We have therefore included terms for heavy leptons and either left- or right-handed neutrinos in Equation 2.8. Possible astrophysical effects of such particles are discussed briefly below.

For neutral-current processes involving nucleons as targets, it is sufficiently general to write the hadronic current as a sum of isoscalar and isovector terms

$$h_\lambda^Z = a_0 V_\lambda^{I=0} + b_0 A_\lambda^{I=0} + a_1 V_\lambda^{I=1}, \quad \text{and} \quad b_1 A_\lambda^{I=1}, \quad 2.9$$

for which the phenomenological parameterization is

$$\begin{aligned} V_\lambda^{I=0} &= \bar{p} \gamma_\lambda p + \bar{n} \gamma_\lambda n, \\ A_\lambda^{I=0} &= \bar{p} \gamma_\lambda \gamma_5 p + \bar{n} \gamma_\lambda \gamma_5 n, \\ V_\lambda^{I=1} &= \frac{1}{2} (\bar{p} \gamma_\lambda p - \bar{n} \gamma_\lambda n), \\ A_\lambda^{I=1} &= \frac{1}{2} g_A (\bar{p} \gamma_\lambda \gamma_5 p - \bar{n} \gamma_\lambda \gamma_5 n). \end{aligned} \quad 2.10$$

The general expressions for l_λ^Z and h_λ^Z contain nine free parameters. Although in principle these can be determined from neutral-current experiments, it is much too

difficult to do this at present. One virtue of the Weinberg-Salam model is that there is only one free parameter, so that experimental determination is much simpler. The various general parameters take the values

$$\begin{aligned}
 c_\mu &= 1, \\
 a_e &= a_\mu = -\frac{1}{2}(1 - 4 \sin^2 \theta_W), \\
 b_e &= b_\mu = \frac{1}{2}, \\
 a_0 &= -\sin^2 \theta_W, \\
 b_0 &= 0, \\
 a_1 &= 1 - 2 \sin^2 \theta_W, \\
 b_1 &= -1.
 \end{aligned} \tag{2.11}$$

Note that μ - e universality is assumed in the W-S model and that h^Z of Equation 2.9 can also be written as

$$h_\lambda^Z = \mathcal{J}_\lambda^3 - 2 \sin^2 \theta_W J_\lambda^{\text{em}} \tag{2.12}$$

where $\mathcal{J}_\lambda^3$ is the V-A isospin current in the same multiplet as h_λ^W in Equation 2.4 or 2.6, and J_λ^{em} is the electromagnetic current. Note that baryon number B , strangeness S , hypercharge Y , electric charge Q , and third component of isospin I_3 are related according to $Y = S + B$ and $Q = I_3 + \frac{1}{2}Y$. These relations facilitate the phenomenological parameterization of h_λ^Z in Equation 2.12, leading to expressions analogous to Equations 2.9 and 2.10, with the particular parameters of the Weinberg-Salam model.

The neutrino-electron-scattering process $\nu_e + e^- \rightarrow \nu_e + e^-$ (and processes such as pair annihilation $e^+ + e^- \rightarrow \nu_e + \bar{\nu}_e$) receive contributions from both charged and neutral currents, as is indicated in the Feynman diagrams of Figure 4. The amplitude is usually parameterized as

$$\frac{iG}{(2)^{3/2} \pi^2} \bar{u}(\nu') \gamma^\mu (1 - \gamma_5) u(\nu) \bar{u}(e') \gamma_\mu [C_V - \gamma_5 C_A] u(e).$$

Then W exchange alone (i.e. pure charged current) gives

$$C_V = C_A = 1,$$

while the Weinberg-Salam model predicts

$$C_V = 2 \sin^2 \theta_W + \frac{1}{2} \quad C_A = \frac{1}{2},$$

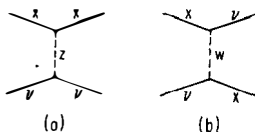


Figure 4 General Feynman diagrams for neutrino scattering: for $e^- + \nu \rightarrow e^- + \nu$, diagrams (a) and (b) are added. For $\nu + p$ (nor $\bar{\nu} + p$), only diagram (a) contributes.

and in the general parameterization of Equation 2.8,

$$C_V = a_e + 1,$$

$$C_A = 1 - b_e.$$

We now discuss coherent nuclear scattering of neutrinos, which was calculated by Freedman (33) for spinless ($J = 0$) nuclei with $Z = N$. In this case, only the $I = 0$ vector current contributes, and the amplitude of Section 1.2 is

$$F(k', k) \sim G \exp(-q^2 R^2) A a_0 \bar{u}_i(v') \gamma^0 u_i(v), \quad 2.13.$$

where an exponential nuclear form factor is assumed, as discussed in Section 1.2. For nuclei with non-zero spin ($J \neq 0$) and $Z \neq N$, one must sum over nucleons, and one then finds, from Equations 2.9 and 2.10,

$$F(k', k) \sim G \exp(-q^2 R^2) [a_0(Z + N) + a_1 \frac{1}{2}(Z - N) + (b_0 + \frac{1}{2}b_1 g_v)(Z \uparrow - Z \downarrow) + (b_0 - \frac{1}{2}b_1 g_A)(N \uparrow - N \downarrow)] \bar{u}(v') \gamma^0 u(v), \quad 2.14.$$

where the arrow indicates spin-up and spin-down nucleons that scatter via the axial neutral current with opposite sign. For all except very light nuclei, the most important term in Equation 2.14 is the term involving $a_0(Z + N)$, so that Equation 2.13 is generally approximately valid.

Nuclear coherent scattering is therefore mainly dependent on the coefficient a_0 of the isoscalar polar-vector part of the weak neutral current.

Nuclear coherent scattering of neutrinos has been considered by Adler (78) in a model (called model A) in which the weak neutral current is a spatial scalar. The effective Lagrangian is simply

$$\mathcal{L}_A = (2)^{1/2} G \bar{\nu} \nu J_s,$$

where J_s is a postulated scalar hadronic neutral current. Astrophysical consequences are discussed below.

2.3 Implications of Present Experiments

All neutrino experiments are notoriously difficult because of the low interaction rate of neutrinos, and associated background and systematic problems. Neutral-current experiments are particularly difficult because the final-state neutrino cannot be detected directly. As a result, present experimental support for any particular model of the weak neutral current is inconclusive. Nevertheless, high-energy accelerator experiments are generally consistent with the Weinberg-Salam model for a value of $\sin^2 \theta_w$ between 0.3 and 0.4. This is true of deep inelastic inclusive reactions at CERN (1-2) and FNAL (3, 4, 6). Furthermore (18), the momentum-transfer distributions in elastic scattering $\nu_\mu + p \rightarrow \nu_\mu + p$ are in good agreement and yield $\sin^2 \theta_w < 0.35$.

An interesting experimental measurement of antineutrino scattering from electrons $\bar{\nu}_e + e \rightarrow \bar{\nu}_e + e$ by Reines and coworkers has been recently reported (109). This experiment is performed with very-low-energy antineutrinos at the Savannah River reactor. The total cross section is consistent both with the pure V-A charged-

current theory and with the Weinberg-Salam model for $\sin^2 \theta_W = 0.29 \pm 0.05$. A more striking comparison emerges when the cross section is plotted against recoil electron energy. The distribution differs from pure V-A and is in qualitative agreement with the W-S model. It is worth emphasizing that this is the first laboratory experimental evidence for neutral-current effects involving electron-type neutrinos, and thus is particularly significant for astrophysics. Accelerator experiments involve only ν_μ and $\bar{\nu}_\mu$, and the untested assumption of universality ($C_\mu = 1$) is generally made in order to calculate ν_e and $\bar{\nu}_e$ processes of astrophysical interest.

In principle, parity-violating effects in atomic physics occur because of the intrinsic parity violation in weak interactions. It was pointed out (112) that with a neutral current, such effects would be enhanced in heavy atoms, due essentially to the nuclear coherence, and that there would be amplitudes of order $G b_e [a_0(Z+N) + \frac{1}{2}a_1(Z-N)]$ for the mixing of atomic levels of opposite parity. Preliminary results (95) of experiments at Oxford University and the University of Washington, on two separate atomic transitions in bismuth, indicate no parity-violating effect with an upper limit of a factor of 3 or 4 below the Weinberg-Salam model prediction. This result provides the first serious challenge for the W-S model, and confirmation by experiments in other atoms and further study is awaited. The results are also relevant for the astrophysical effectiveness of nuclear coherent scattering because a strong limit on the product $b_e a_0$ is implied by the present negative results on parity violation. It is too early to conclude that a_0 is small. Indeed, theoretical models (113, 114) have been constructed that predict no atomic parity violation, but appreciable coherent scattering.

Uncertainty about the nature of the weak neutral current poses a dilemma for the astrophysicist. The weak interaction is considered fundamental in determining the outcome of the late stages of stellar evolution. The weak neutral current seems to be particularly important for stellar collapse and supernova explosions. Which model should be used? The Weinberg-Salam model is simple and elegant, but it may be incomplete at best. Much more experimental data is needed before the parameters of the weak neutral current can be determined. Information on the space-time structure and, in particular, on the isoscalar component of the neutral current are very much needed and anxiously awaited.

2.4 *Neutrino Interactions in Stellar Matter*

A physicist working at CERN, FNAL, or SLAC would rapidly develop ulcers were the laboratory conditions those of the stellar interior. In STAR, neither the target nor the projectile are confined to beams; instead, the local motions of the particles are for the most part random, with a wide variation of kinetic energies. Matter densities are so high that the less massive particles become packed nearly one atop the other, and particles are continuously created and destroyed by the thermal bath.

Nature quite successfully handles the situation. You and I are living proof—the stuff of the stars. Astrophysicists, though not as clever as Nature, are nevertheless stubborn and curious. They persist in the effort to understand how all of those precious elements like carbon and oxygen are released into a universe full of hydrogen. With many neutrinos and several theories, several of which are discussed in

Section 2.2 and none of which have been so well tested that we should stake our lives on them, the astrophysicist might come close to the answer.

The discussion that follows involves general results of theories that include only vector and axial vector neutral currents (e.g. gauge theories). When the calculations of pertinent processes have been done in other weak-interaction models, those results are compared with the predictions of the Weinberg-Salam gauge theory.

Electrons, and positrons if the temperature is not too low or the density too high, are in thermal equilibrium. (Temperatures in the region of interest are generally insufficient to thermally produce muons.) The electromagnetic interactions are able to maintain this equilibrium on time scales much shorter than those of the weak interactions. Thus the electrons and positrons that enter into the neutrino processes may be described by Fermi-Dirac distributions. In neutral matter, the proton-number density, matter density, average number of protons per amu (Y_e), and electron- and positron-number densities are related [cf (81)] by

$$n_p = N_A \rho Y_e = n_{e^-} - n_{e^+}. \quad 2.15.$$

When the electrons are degenerate ($\mu_{e^-}/kT \gg 1$), $n_{e^+} \ll n_{e^-}$. By the time nuclear photodisintegration and collapse have been initiated, the electrons will be degenerate; the degree of degeneracy depends upon how advanced the collapse is (47, 77, 82).

The nucleons, on the other hand, will become noticeably degenerate only at advanced stages of collapse. The chemical potential of the electrons will be sufficiently high that the average electron will be relativistic ($p^2 \simeq E^2$), whereas the nucleons can be described by a nonrelativistic ($p^2 \simeq 2mE$) Maxwellian distribution. The lepton energies ($\sim$ tens of MeV) will still be low enough compared with the nucleon and nuclear rest masses that, in neutrino processes involving nucleons and nuclei, the calculation may be performed in the target rest frame (which is approximately the fluid rest frame). Yueh & Buchler (83) have included the Maxwellian distribution in their treatment of the transport equation.

One effect of the stellar matter on the single-particle processes is thus to inhibit those interactions with electrons in the final state. A factor of

$$1 - f_{e^-}(E) = 1 - [\exp(E - \mu_{e^-})/kT + 1]^{-1}$$

appears in the differential interaction rate for final-state electrons, and a factor of $f_{e^-}(E)$ appears for initial-state electrons. The single-particle interactions become statistical averages, the interaction probabilities being weighted by the product of available and filled momentum space.

Consider the β decay processes, which only proceed by way of charged currents. In the absorption of ν_e , the proton is essentially produced at rest. The cross section for $\nu_e + n \rightarrow p + e^-$ in stellar matter is just the single-particle cross section multiplied by the inhibition factor $1 - f_{e^-}(\omega + \Delta m)$ (84, 34),

$$\begin{aligned} \sigma_{\nu_e n \rightarrow p e^-} &= \sigma_0 \left[\frac{1}{2}(\alpha^2 + 1) + \frac{1}{4}(\alpha^2 - 1) \right] \omega^2 (\omega + \Delta m) \\ &\quad \times [\omega^2 + 2\omega\Delta m + \Delta m^2 - 1]^{1/2} [1 - f_{e^-}(\omega + \Delta m)], \end{aligned} \quad 2.16.$$

$$\sigma_0 = 1.721 \times 10^{-44} \text{ cm}^2.$$

· Absorption of $\bar{\nu}_e$ is uninhibited, the final state containing a positron:

$$\sigma_{\bar{\nu}_e p} \rightarrow n e^+ = \sigma_0 \left[\frac{1}{2}(\alpha^2 + 1) + \frac{1}{4}(\alpha^2 - 1) \right] \omega^2 \cdot (\omega - \Delta m) [\omega^2 - 2\omega\Delta m + \Delta m^2 - 1]. \quad 2.17.$$

A simple argument of detailed balance applied to the single-particle rates, or direct calculation from the matrix element (83, 85) yields the differential rate of production of neutrinos of energy ω' , for the electron-capture process:

$$\begin{aligned} dR_{ep} \rightarrow n \nu_e &= R_0 \left(\frac{2n_p^f}{\pi} \right) \left[\frac{1}{2}(\alpha^2 + 1) + \frac{1}{4}(\alpha^2 - 1) \right] \omega'^2 (\omega' + \Delta m) \\ &\times [\omega'^2 + 2\Delta m \omega' + \Delta m^2 - 1]^{1/2} f_e(\omega' + \Delta m) d\omega' d\Omega', \end{aligned} \quad 2.18a.$$

$$R_0 = \frac{2G^2 m_e^5 c^2}{(2\pi)^3 \hbar} = 5.6739 \times 10^{-5} \text{ sec}^{-1}. \quad 2.18b.$$

The dimensions of $R_{ep} \rightarrow n \nu_e$ are $\text{cm}^{-3} \text{ sec}^{-1}$; and n_p^f is the density of free protons, not to be confused with Equation 2.15. Note that Equation 2.18 yields the energy spectrum of the electron-capture neutrino, and that the emissivity $d\mathcal{E}_{ep} \rightarrow n \nu_e$ is obtained by multiplying Equation 2.18 by $\omega(m_e c^2)$. Approximate expressions for $R_{ep} \rightarrow n \nu_e$ and $\mathcal{E}_{pe} \rightarrow n \nu_e$ when Δm can be neglected are given by Bludman & Van Riper (86).

If the electron gas has a dielectric constant less than unity, the photon can acquire a small mass, and thereby decay. In fact, the plasmon process, $\Gamma \rightarrow \nu + \bar{\nu}$, is a stellar parasite, robbing the star of energy, but unable to survive without its host.

The plasmon decay, pair annihilation, and photoproduction neutrino-pair emissivities ($\text{erg sec}^{-1} \text{ cm}^{-3}$) have been calculated by Dicus (32), in the context of the Weinberg-Salam model. For present values of the Weinberg angle, the ratios of emissivities in the W -S model to the W -meson theory are $\lesssim 2$. Using arbitrary vector and axial vector couplings, Dicus's results may be extended to include heavy-lepton neutrino production. General expressions for the emissivities are:

$$\begin{aligned} \mathcal{E}_{\text{pair}} &= [7(C_V^e{}^2 + C_V^{\mu}{}^2 + C_V^L{}^2) - 2(C_A^e{}^2 + C_A^{\mu}{}^2 + C_A^L{}^2)]P_1 \\ &+ 9(C_V^e{}^2 + C_V^{\mu}{}^2 + C_V^L{}^2)P_2 \end{aligned} \quad 2.19a.$$

$$\begin{aligned} &+ (C_V^e{}^2 + C_V^{\mu}{}^2 + C_V^L{}^2 + C_A^e{}^2 + C_A^{\mu}{}^2 + C_A^L{}^2)P_3, \\ \mathcal{E}_\Gamma &= (C_V^e{}^2 + C_V^{\mu}{}^2 + C_V^L{}^2)F, \end{aligned} \quad 2.19b.$$

$$\begin{aligned} \mathcal{E}_\gamma &= (C_V^e{}^2 + C_V^{\mu}{}^2 + C_V^L{}^2 + C_A^e{}^2 + C_A^{\mu}{}^2 + C_A^L{}^2)G_1 \\ &+ (C_V^e{}^2 + C_V^{\mu}{}^2 + C_V^L{}^2 - C_A^e{}^2 - C_A^{\mu}{}^2 - C_A^L{}^2)G_2. \end{aligned} \quad 2.19c.$$

The emission rates, $R(\text{sec}^{-1} \text{ cm}^{-3})$, for these interactions follow analogous formulae. Both $\mathcal{E}$ and R have been numerically determined for general C_V^i and $C_A^i (i = e, \mu, L)$, over a wide range of densities and for temperatures up to several tens of MeV (87). If the arbitrary couplings are interpreted in the WS model, and the additional assumption of universality is made, then

$$C_V^e = \frac{1}{2} + 2 \sin^2 \Theta_w \simeq 1.3, \quad C_A^e = \frac{1}{2},$$

$$C_V^e = C_V^L = (1 - C_V^e),$$

and

$$C_A^e = C_A^L = (1 - C_A^e).$$

The ratios of the rates in this model of neutral currents to the charged-current theory, as functions of the number of μ and L neutrino types, J ($J = 2$ for 1 heavy-lepton neutrino), are approximately $(0.845 + 0.145 J)$ for the pair and photo rates, and $(1.44 + 0.04 J)$ for the plasma rates. The branching ratios for $\nu_J \bar{\nu}_J$ to $\nu_e \bar{\nu}_e$ in the theory with neutral currents are

$$\left(\frac{\nu_J \bar{\nu}_J}{\nu_e \bar{\nu}_e} \right)_{\text{pair}, \gamma} \simeq 0.172,$$

$$\left(\frac{\nu_J \bar{\nu}_J}{\nu_e \bar{\nu}_e} \right)_{\Gamma} \simeq 0.028.$$

The electron bremsstrahlung process, $e^- + Z \rightarrow e^- + Z + \nu + \bar{\nu}$, will only be important when the electron degeneracy is extreme and the thermal processes above have shut off. The result of a calculation that includes general neutral currents (37) is that the charged-current emissivity is enhanced by only a factor of ~ 1.31 when ν_e and $\bar{\nu}_e$ pairs are produced, and by a factor of ~ 1.52 when a single ν_e pair is also included ($C_V^e \simeq 1.4$, $C_A^e = \frac{1}{2}$). The branching ratio $\nu_J \bar{\nu}_J / \nu_e \bar{\nu}_e$ is ~ 0.186 .

Neutral currents thus seem to have little effect on the charged-current neutrino-production mechanisms, and electron-neutrino pairs are the dominant neutrino type produced. In actual collapse situations, comparison of the emissivities shows (82, 88, 89) that electron capture is the dominant neutrino source; at later stages of the collapse, antineutrinos can become important, but still considerably less so than neutrinos (77); however, at late times, when the core is quite opaque to neutrinos, Wilson (77) has found that muon neutrinos may become the dominant luminosity source, due to their smaller electron-scattering cross sections.

Tensor, scalar, and pseudoscalar neutral currents have also been included in the basic neutrino-luminosity sources, and we briefly note the results. Based upon the $\Gamma \rightarrow \nu + \bar{\nu}$ process and a comparison of degenerate-dwarf cooling times with observations of hot white dwarfs, Sutherland et al (90) have deduced that an upper limit on the neutral-tensor coupling constant is $C_T \lesssim 0.06$. Dicus & Kolb (91), in a study of the pair annihilation, photoproduction, bremsstrahlung, and $\gamma + \gamma \rightarrow \nu + \bar{\nu}$ processes, have found less restrictive bounds, based upon a similar comparison with observed stellar lifetimes: $C_T \lesssim 3.6$, $C_S \lesssim 16$, $C_P \lesssim 14$.

The general conclusions of the effect of neutral currents on the emissivities hold likewise for the neutrino-electron scattering interaction (34). In terms of arbitrary vector and axial vector couplings, the reaction rates and cross sections follow the general form (for electron-type neutrinos)

$$R_{e\nu} \rightarrow e\nu = (C_V + C_A)^2 R_1 + (C_V - C_A)^2 R_2 + (C_A^2 - C_V^2) R_3. \quad 2.20.$$

For relativistic electrons, $R_3 \rightarrow 0$. Muon neutrinos can now scatter with electrons, by the exchange of a neutral meson, but the rates are strongly suppressed, being at best

$\sim 17\%$ of the ν_e scattering rate. The overall effect of the neutral current is to decrease the $R_e \nu_e$ rate by $\sim 20\%$ from the rate for charged currents. Table 5 gives some values for $(\nu_e e)$ scattering rates.

The great importance of neutral currents to astrophysics is not that they alter existing charged current processes, but that neutral currents permit new interactions that may have dramatic effects in the stellar matter.

Within the neutron star, neutron bremsstrahlung ($n + n \rightarrow n + n + \nu + \bar{\nu}$) may play as important a role in cooling by neutrino loss (118) as does the modified URCA process. While in white dwarfs with central temperature $\lesssim 10^8 K$ (at the high temperatures of gravitational collapse, the purely leptonic processes will dominate), semileptonic cooling by nuclear de-excitation to neutrinos ($A^* \rightarrow A + \nu + \bar{\nu}$) may compete with the purely leptonic cooling mechanisms (36). Both processes involve V and A neutral currents only, and with the assumption of universality, both produce e , μ , and L neutrinos with equal probability.

In the collapsing stellar core, purely neutral-current processes significantly increase the opacity of the matter to neutrinos—not only in the mantle of heavy nuclei at the exterior of the core, but deep within the core as well, where the nuclei are nearly fully dissociated into free nucleons. The possible implications of these neutrino processes is drawn in Section 3. The discussion that follows is concerned with providing estimates and comparison of the sources of neutrino opacity. Because the degeneracy (if it is present at all) of the nucleons and nuclei may be neglected, target-inhibition factors are absent in the cross sections. Furthermore, because $\omega \ll m_n, m_p, m_A$ is a good approximation for neutrinos produced in the star, the cross sections are calculated in the c.m. frame (fluid rest frame), and thus are conservative ($\omega' \simeq \omega$) and proportional to ω^2 .

We have already seen how neutrino scattering at small momentum transfer on spinless, isoscalar nuclei is greatly enhanced, due to the simple principle of coherent superposition of amplitudes. In the model of Freedman (33), the formulae for the cross sections (with z the cosine of the scattering angle) are:

$$\frac{d\sigma_A}{dz} \simeq \frac{1}{8} a_0^2 \sigma_0 A^2 \omega^2 (1+z), \quad 2.21a.$$

$$\sigma_A \simeq \frac{1}{4} a_0^2 \sigma_0 A^2 \omega^2, \quad 2.21b.$$

where it is assumed that $\omega \ll m_A$ (the nucleus is pointlike). The application of this model to $Z \neq N$ spinless nuclei (34) introduces a coupling to the nuclear isospin

$$[I_3 = \frac{1}{2}(Z - N)]:$$

$$\frac{d\sigma_{Z,N}^{(J=0)}}{dz} \simeq \frac{\sigma_0}{8} a_0^2 A^2 \left[1 + \frac{1}{2} \frac{a_1}{a_0} \frac{Z - N}{A} \right]^2 \omega^2 (1+z), \quad 2.22a.$$

$$\sigma_{Z,N}^{(J=0)} \simeq \frac{\sigma_0}{4} a_0^2 A^2 \left[1 + \frac{1}{2} \frac{a_1}{a_0} \frac{Z - N}{A} \right]^2 \omega^2. \quad 2.22b.$$

In the W-S theory, $a_1 = 1 + 2a_0 = 1 - 2 \sin^2 \Theta_w$.

Table 5 Neutrino-electron scattering rates (sec^{-1} per neutrino)^a

$T/10^9$	$\frac{\mu_e^a}{m_e c^2}$	$\rho Y_e (\text{g cm}^{-3})$	$\Omega = 4 \text{ MeV}$			$\Omega = 8 \text{ MeV}$			$\Omega = 16 \text{ MeV}$			$\Omega = 24 \text{ MeV}$		
			R_1	R_2	R_3	R_1	R_2	R_3	R_1	R_2	R_3	R_1	R_2	R_3
10	10	1.2+9	1.9+0	7.4-1	8.0-3	6.6+0	1.9+0	1.8-2	1.6+1	4.7+0	2.2-2	2.6+1	7.7+0	2.3-2
	25	1.6+10	9.1+0	5.8+0	9.2-3	4.3+1	1.9+1	4.0-2	2.8+2	7.0+1	1.9-1	5.5+2	1.4+2	2.5-1
	50	1.2+11	3.2+1	2.6+1	9.2-3	1.4+2	9.5+1	4.0-2	9.9+2	4.4+2	2.5-1	3.6+3	1.1+3	8.1-1
25	10	2.7+9	1.1+1	4.1+0	2.6-2	2.6+1	8.9+0	3.8-2	6.2+1	1.9+1	3.1-1	9.7+1	3.0+1	5.0-2
	25	1.9+10	4.2+1	2.3+1	4.4-2	1.2+2	5.4+1	9.2-2	4.5+2	1.4+2	3.1-1	8.5+2	2.4+2	3.1-1
	50	1.3+11	1.4+2	9.9+1	4.6-2	4.0+2	2.5+2	1.0-1	1.7+3	7.5+2	4.5-1	4.8+3	1.5+3	9.3-1
50	10	7.8+9	7.2+1	2.8+1	1.0-1	1.6+2	5.7+1	1.2-1	3.6+2	1.2+2	3.3-1	5.6+2	1.8+2	1.6-1
	25	3.2+10	2.0+2	9.0+1	1.8-1	4.7+2	1.9+2	2.5-1	1.2+3	4.2+2	4.0-1	2.1+3	6.7+2	5.0-1
	50	1.6+11	5.6+2	3.3+2	2.4-1	1.4+3	7.3+2	3.5-1	4.0+3	1.7+3	7.4-1	8.5+3	3.0+3	1.3+0
100	10	2.8+10	7.2+2	2.7+2	5.3-1	1.5+3	5.5+2	5.8-1	3.2+3	1.1+3	6.6-1	5.1+3	1.7+3	7.2-1
	25	8.3+10	1.3+3	5.4+2	8.2-1	2.8+3	1.1+3	9.3-1	6.3+3	2.3+3	1.1+0	1.0+4	3.5+3	1.3+0
	50	2.6+11	2.9+3	1.4+3	1.2+0	6.3+3	2.9+3	1.5+0	1.5+4	6.1+3	2.0+0	2.6+4	9.6+3	2.6+0

^aThe first number is the mantissa; the second is the power of ten.

The scalar neutral current model of Adler (model A) produces a different angular dependence in the differential cross section, as well as a different interaction strength:

$$\frac{d\sigma_A}{dz} = \frac{\sigma_0}{8} \mathcal{A}_s^2 A^2 \omega^2 (1-z) \quad 2.23a.$$

$$\sigma_A = \frac{\sigma_0}{4} \mathcal{A}_s^2 A^2 \omega^2. \quad 2.23b.$$

The angular difference is understood as follows: in coherent transitions when the nucleus angular momentum remains unchanged, the scalar interaction flips the neutrino helicity and must vanish in the forward direction, whereas the vector interaction does not alter the helicity and must vanish in the backward direction. An upper limit to the magnitude of the scalar coupling is $|\mathcal{A}_0|^2 \lesssim 4$ (78). The main effect of this different angular dependence is to increase the neutrino radiation pressure.

The speculative model of De Rujula, Georgi & Glashow (DG²) (79), which includes an additional V + A neutral current, has been discussed and compared with the WS theory by Bernabeu (35). The results have been given in terms of the cross section for momentum transfer,

$$\Sigma = \int d\Omega (1-z) \frac{d\sigma}{d\Omega},$$

for general nuclei with nonzero spin. Coherent scattering is modified by an isospin term T (which agrees with Equation 2.16), and a correction, $\mathcal{S}$, which contains both spin and isospin effects. In DG², the $\mathcal{S}$ correction is absent, and in WS it depends upon the particular nuclear model chosen. Bernabeu has expressed $\mathcal{S}^2$ in terms of the f_t value for the β^+ decay of an isospin state $(I, I) \rightarrow (I, I-1)$. The resulting formulae are

$$\Sigma = \frac{1}{6} \sigma_0 \eta_0^2 A^2 \omega^2 [(1+T)^2 + \mathcal{S}^2], \quad 2.24a.$$

$$T = \frac{Z-N}{\eta_0 A} \begin{cases} \frac{1}{2}(1+2a_0) & \text{WS,} \\ 1+\alpha_0 & \text{DG}^2 \end{cases} \quad 2.24b.$$

$$\mathcal{S}^2 = \frac{5}{12a_0^2 A^2} (N-Z) \left[\frac{K}{f_t} + (Z-N) \right], \quad 2.24c.$$

$$K = 2\pi^3 \ln \left(\frac{2h}{G^2 m_e^5 c^2} \right), \quad 2.24d.$$

where η_0 is equal to a_0 for WS, and α_0 for DG². Some of Bernabeu's results for nuclei near Fe are presented in Table 6. The first row under each species is for the WS model ($a_0^2 = 0.16$); row two is DG² ($\alpha_0^2 = 0.49$). General conclusions to be drawn are that for $A \geq 10$, modifications to coherent scattering $\lesssim 10\%$, independent of the model. For $A > 10$, the ratio of cross sections $\Sigma(\text{DG}^2)/\Sigma(\text{WS}) \simeq 3.1$, while the same ratio for neutrons is 0.45. Thus, the DG² model is a more effective means of transferring momentum from the core to the mantle than is the WS theory; that is, in the core,

Table 6 Spin and isospin corrections to coherent scattering

Nucleus	$(1 + T)^2$	S^2
Mn ⁵⁵	1.05	0
	1.08	
Fe ⁵⁶	1.04	0
	1.06	
Ni ⁵⁸	1.02	0
	1.03	
Co ⁵⁹	1.04	0
	1.07	
Ni ⁶⁰	1.03	0
	1.06	

λ_{eff} is larger in DG² than in WS, while in the mantle, λ_{eff} is less in DG² than in WS. Observe that for equal coupling strengths, $|a_0| = |\mathcal{A}_0|$, the scalar neutral current is twice as effective as the V-A neutral current in terms of momentum transfer in the $A > 1$ mantle. Results of Wilson's collapse calculations tend to reflect this model-dependent gradient of effectiveness in momentum transport.

Within the center of the star, the principal semileptonic neutrino opacities arise from elastic scattering with nucleons. With the added contribution from the axial vector current, the cross sections are significantly greater than those of coherent scattering for $A = 1$. At low momentum transfer, the interactions are (34):

$$\sigma_{vp} = \frac{1}{4}\sigma_0[(C_V - 1)^2 + 3(C_A - 1)^2]\omega^2, \quad 2.25.$$

$$\sigma_{vn} = \frac{1}{4}\sigma_0\omega^2. \quad 2.26.$$

Due to the assumption of a neutral current of the form of Equation 2.5a, the cross section for neutron scattering is independent of the Weinberg angle when neutron-structure effects are ignored. The contribution to scattering comes solely from the isovector V-A current (Equation 2.12). The proton interaction had previously been calculated by Weinberg (92), and the general result (2.24) of Bernabeu, which includes both nucleon cross sections, shows that

$$\Sigma_{A=p}(\text{WS})/\Sigma_{A=p}(\text{DG}^2) \simeq 13 \quad (\mathcal{G}^2 = 12.18).$$

A further correction to coherent scattering arises from the ion-ion correlations of nuclei in the stellar core (cf Ref. 93). The effective cross section is obtained from the liquid structure factor $S(\Delta a)$:

$$\frac{d\sigma_{\text{eff}}}{d\Omega} = \frac{d\sigma}{d\Omega} S(\Delta a),$$

where

$$\Delta = \frac{\omega}{c}[2(1-z)]^{1/2}$$

is the momentum transfer, and a is a measure of the mean spacing of the nuclei,

$$a = \left(\frac{4\pi}{3} \frac{N_A \rho}{A} \right)^{-1/3}$$

However, for conditions within the collapsing core, and for values of the typical neutrino energies encountered (~ 20 MeV), the suppression by the structure factor is estimated to be only ~ 10 – 15% . At neutron-star densities, when the neutrino wavelength encompasses several neutrons, the resulting multiple scattering would be like an index-of-refraction effect by the continuous neutron media, rather than coherent scattering from point-scattering centers. Correlation effects from such high-density scattering are also unimportant.

Finally, we must mention a bugaboo that we have tactfully ignored up to this point. What happens when densities become high enough that all of these wonderful cross sections cause a significant buildup of the neutrino number? At some time in the evolution of stellar collapse, in some locations of the core, and at some neutrino energies, the filling of neutrino phase space must be included in all of the above neutrino processes (cf Ref. 94). That is, a factor of

$$1 - f_\nu(\mathbf{r}, \mathbf{q}, t)$$

must be inserted into each differential interaction rate that has a neutrino in the final state. The general neutrino-distribution function depends upon seven parameters, although symmetries of the transport problem make some coordinates ignorable.

In addition to requiring inhibition factors in the rates, when the neutrino number becomes sufficiently high that they begin to scatter elastically with each other, they may do so with cross sections that are comparable to those of neutrino-electron scattering (80). This process, like the semileptonic interactions that have just been discussed, depends solely on the exchange of neutral currents.

In order to provide some feel for the size of typical mean free paths and the relative importance of the neutrino opacities, we have included in Table 7 the approximate data for ν_e in the Weinberg-Salam model ($C_V = 1.3$, $C_A = 0.5$). The table is loosely based upon an actual collapse configuration (88, 89). The local physical conditions should only be considered roughly representative of two stages in a stellar collapse. The situation is not simply a homogeneous mixture of electrons, free nucleons, and a single nuclear species as heavy as iron. The number densities are those for unbound particles; that is, absorption of neutrinos within a nucleus has been ignored. At the temperatures and densities indicated in Table 7, the number of positrons is negligible with respect to the number of electrons; therefore, $n_e \simeq N_A \rho Y_e$, the total number density of protons, both free and bound. For a given interaction, the mean free path can vary considerably—over several orders of magnitude—with the incident neutrino energy. The mean free paths show similar variation in magnitude from interaction to interaction.

Before leaving the weak-interaction details, let us reiterate that the weak neutral current can only cause dramatic effects when it allows a process to go that would otherwise be forbidden, as in the case of coherent scattering. Otherwise, the effects tend to be small ($\lesssim$ factor of 2 change in the rate).

Table 7 Electron-neutrino mean free paths^a

Local physical conditions	mfp (cm)	$\omega = 4$ MeV	$\omega = 8$ MeV	$\omega = 16$ MeV	$\omega = 24$ MeV
$\rho/10^9 = 10 \text{ g cm}^{-3}$					
$T/10^9 = 10 \text{ K}$	λ_e	2.8+8	6.5+7	9.6+6	2.7+6
$X_A = 0.80$					
$X_p = 0.093$	λ_n	5.9+9	1.5+9	3.7+8	1.6+8
$X_n = 0.107$					
$Y_e = 0.464$	λ_p	8.1+9	2.0+9	5.0+8	2.2+8
$n_A/10^{31} = 8.60 \text{ cm}^{-3}$					
$n_p/10^{31} = 55.93 \text{ cm}^{-3}$	λ_A	8.5+7	2.1+7	5.3+6	2.4+6
$n_n/10^{31} = 64.53 \text{ cm}^{-3}$					
$n_e/10^{31} = 279.47 \text{ cm}^{-3}$	λ_{ab}	4.2+9	1.0+9	2.6+8	5.6+7
$\mu_e/m_e c^2 = 46.54$					
$A = \text{Fe}^{56}$					
$\rho/10^9 = 200 \text{ g cm}^{-3}$					
$T/10^9 = 27.14 \text{ K}$	λ_e	7.8+7	2.9+7	7.9+6	3.2+6
$X_A = 0.50$					
$X_p = 0.050$	λ_n	7.0+7	1.8+7	4.4+6	1.9+6
$X_n = 0.450$					
$Y_e = 0.282$	λ_p	7.5+8	1.9+8	4.7+8	2.1+7
$n_A/10^{31} = 107.55 \text{ cm}^{-3}$					
$n_p/10^{31} = 602.30 \text{ cm}^{-3}$	λ_A	6.8+6	1.7+6	4.2+5	1.9+5
$n_n/10^{31} = 5420.7 \text{ cm}^{-3}$					
$n_e/10^{31} = 3397.0 \text{ cm}^{-3}$	λ_{ab}	5.0+7	1.2+7	2.2+6	6.5+5
$\mu_e/m_e c^2 = 35.41$					
$A = \text{Fe}^{56}$					

^a Same representation as Table 4. Mean free paths for ν_e in the Weinberg-Salam model ($\sin^2 \theta_w = 0.40$, $C_A = \frac{1}{2}$), as functions of incident neutrino energy (ω in MeV). Subscripts on λ refer to scattering with electrons, neutrons, protons, and nucleus A ; and absorption on neutrons. Local physical conditions are approximate. Effects of neutrino degeneracy have been ignored.

3 COLLAPSE AND SUPERNOVA EXPLOSIONS

Let us now examine how these weak processes effect the supernova-gravitational collapse scenario. Other applications, such as the cooling of neutron stars and the effects on stellar evolution, are also important, but are not as potentially dramatic as supernovae.

3.1 Collapse Scenario

As we mentioned in Section 1, all massive stars tend to evolve to the same core configuration—a $1.4 M_\odot$ Fe-Ni core surrounded by burning shells of Si, O, Ne, C, He, and H. This Fe-Ni core begins collapsing on a ρ - T trajectory that passes through $\rho = 10^{10} \text{ g cm}^{-3}$ and $T \sim 10^{10} \text{ K}$, as shown in Figure 5 (88). As the core collapses through a density of around $10^{11} \text{ g cm}^{-3}$, electron captures go at extremely high

rates. Thus, at this point the neutrinos from electron captures dominate, and are regular electron neutrinos. As the collapse continues and the temperature continues to rise, the neutrino pair processes may become important.

Then, because the pair processes go both by a charged and a neutral current, they not only produce electron-neutrino pairs, but also produce muon-neutrino pairs and, if heavy leptons exist, heavy lepton-neutrino pairs in comparable numbers to the muon neutrino pairs. It has been pointed out by Mikaelian (119) and Freedman (97) that this could be extremely important, because it could increase the total neutrino flux.

It seems very clear that neutrinos are produced and are the main way in which the collapsing star loses its energy. However, the question of whether or not these neutrinos are capable of causing mass ejection is much more complicated. The primary neutrino-opacity sources are described in Section 2. The opacity sources divide themselves into those that go via a charged current, or a charged and a neutral current, such as neutrino absorption by neutrons and neutrino-electron scattering, and the pure neutral-current process, which is neutrino scattering from nucleons and nuclei. We have seen that the cross section for neutrino scattering from complex nuclei scales like atomic weight-squared at low energies and thus, it has been

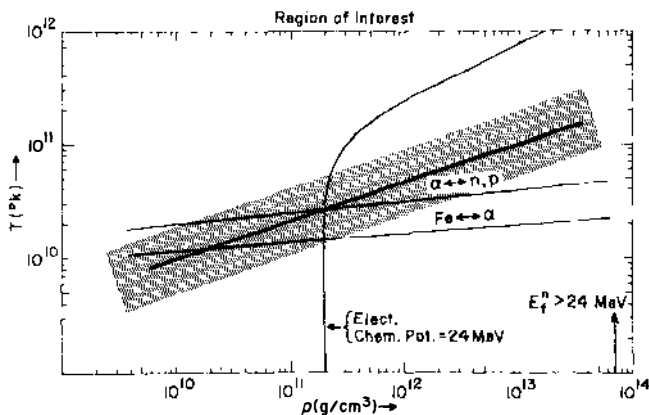


Figure 5 The temperature-density (T - ρ) plane shows the probable region of interest in stellar-collapse calculations. A trajectory with $\rho \propto T$ going through the point $\rho = 10^{10}$ is shown. Cores for stars with $M \geq 8 M_{\odot}$ follow similar trajectories. The region of interest may include the shaded region that surrounds this line. Also illustrated in the figure are the photo-disintegration contours, where 50% of the Fe-Ni has broken up into alphas and where 50% of the alphas have broken up into n and p . In fact, each of these photodisintegration lines turns down at high densities, but the exact location of the turndown depends on poorly known nuclear physics and thus has not been shown. The contour for the chemical potential $\mu = 24$ MeV is shown. For points to the right of this contour, electron capture will occur, even on the most neutron-rich nuclei. Therefore, to the right of $\alpha \sim 2 \times 10^{11}$ is a neutrino source and the average atomic weight $A \sim 1$, whereas to the left of $\sim 2 \times 10^{11}$ and below the $\alpha \rightarrow n, p$ line, $A > 1$ and not as many neutrinos are being produced.

noted, this mechanism may provide a way to eject the heavy element-rich mantle (iron and beyond), while allowing the central, neutronized core to collapse to make a neutron star. This would explain why the outer part of the star is ejected and the inner part implodes to make a very dense remnant. However, there have been a number of complications in this seemingly nice scenario. One important one is the effect of degeneracy (82, 88, 94), both the degeneracy of electrons and that of neutrinos in the very central core. The electron degeneracy causes inhibitions in all reactions that have electrons in the final state, such as the neutrino-electron scattering and the neutron absorption by neutrons going to a proton plus an electron.

The neutrino degeneracy also causes an inhibition in reactions that have neutrinos in the final state, but in completely conservative scattering processes, the effect can cancel (98, 99). That is, the enhancement in the scattering caused by the initial neutrino state driving the reaction in a forward direction balances the inhibition caused by neutrinos in the final state. This cancellation of the inhibition effects, due to the neutrino being both in the forward and backward channel, occurs only when one has conservative scattering; that is, scattering where there is no energy change, or only a very small one, for the neutrino. This is the case for the neutrino scattering off nuclei and nucleons. However, the neutrino-electron scattering is very non-conservative (34). Therefore the cancellation of the degeneracy effects does not occur and the situation is much more complex. In the central-core regions of the collapsing stars, the neutrino scattering from nucleons dominates and the neutrino-degeneracy effects for this region tend to cancel. This cancellation and noncancellation is extremely important because it determines the energy spectrum of the neutrinos emerging from the core, as well as their ability to emerge from the core. Since neutrino cross sections go roughly like E^2 , the energy of the neutrinos coming out of the core is extremely important.

The effect of all of these processes is to make the following of the detailed neutrino transport in the core a very difficult problem. It is currently being attacked by a variety of methods from simple diffusion to flux-limited diffusion, to S_n diffusion by Yueh & Buchler (100, 115), to the Monte Carlo treatment by Tubbs (85) and Tubbs, Schramm, Weaver & Wilson (116). It does seem that the central core has extremely large neutrino opacities, which would make some form of diffusion a very good approximation. However, in the region just outside the central core, where the neutrino mean free paths are comparable to the size of the region, and where the neutrinos must act to produce the supernova (if they do so at all), the transport is extremely complicated. The importance of this region was pointed out by Schramm & Arnett (88). In the past few years, it has been noted that an equally important question is what is the luminosity of the neutrinos coming out of the central core into this mantle region, where the action might take place. It is in this central-core region that neutrino degeneracy seems to be very important, and it is here that the neutrino luminosity is determined.

It has been shown by Bruenn, Arnett & Schramm (39, 65) that the critical question of whether or not the neutrinos are able to drive off the outer material via momentum transport is that of whether or not the neutrino luminosity emerging from the central-core region is greater than the Eddington limit for neutrinos. We can estimate the

Eddington limit luminosity in the following way (39):

$$L_{\nu}^{\text{Edd}} \approx \frac{4\pi GMc}{\kappa_{\nu} \langle 1 - \cos \theta \rangle},$$

where M is the interior mass, κ_{ν} is the neutrino opacity in the shell, and $\langle 1 - \cos \theta \rangle$ is the average scattering angle.

The neutrino opacity in the shell is dominated by the coherent scattering process. If the shell has a composition of iron or heavier metals, it is very easy to estimate its opacity and thus to calculate the neutrino luminosity necessary to push it off. This luminosity is approximately 10^{54} ergs sec^{-1} . The number of neutrinos coming from electron capture alone is of the order of 10^{57} neutrinos, and the typical energy of these neutrinos is of the order of 15–20 MeV. If they were emitted within the free-fall time for a core at 2×10^{11} g cm^{-3} , then the neutrino luminosity would indeed be greater than the Eddington limit. However, we know that they are not emitted over a free-fall time since a core is collapsing somewhat slower than free-fall. In fact, due to the finite diffusion time out of the core, the time scale for which the neutrinos are escaping is quite a bit longer than free-fall. The whole collapse is regulated by the rate at which neutrinos carry away the binding energy. The question is, what is this time scale over which it takes the neutrinos to leave the core? If the neutrinos can get out more

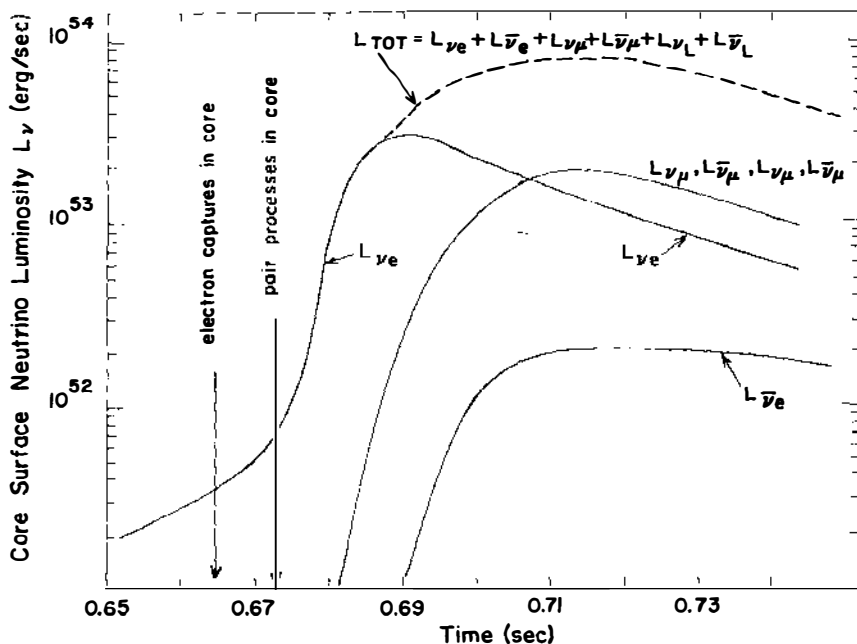


Figure 6 The neutrino luminosity versus time for a typical (though by no means conclusive) gravitational-collapse calculation. This is based on the work of Wilson (101).

rapidly, then in fact we will have a luminosity near the Eddington limit and the neutrinos can cause ejection. However, it should be remembered that even if the neutrinos could stream out of the core, the collapse would not be free-fall, because the electron degeneracy pressure and thermal nucleon pressures are not zero.

In simple hydrodynamic calculations by Wilson (101), it is found that using reasonable, though by no means certain, estimates for cross sections and emission rates, the luminosity of the core falls below the Eddington limit by a factor of approximately 3, thus meaning in these particular calculations that the neutrinos are unable to cause the ejection. One should remember, though, that all of these calculations have large numbers of uncertainties so that the factor of 3 is certainly not fatal at the present stages of the calculations. Thus the questions as to whether or not neutrinos are able to cause an ejection via momentum transport come down to whether or not the neutrino transport is sufficiently rapid to get the neutrinos out of the core with a luminosity greater than the Eddington limit. Another way to augment the luminosity is via neutrinos from heavy leptons, since these neutrinos are produced in equal numbers with the muon neutrinos in the pair processes. The more types of neutrinos there are, the greater the neutrino luminosity would be.

Figure 6 shows the neutrino time spectrum of typical gravitational-collapse calculations. Note that initially there is a burst of the electron neutrinos that gradually diffuse out of the core, followed by the emission of the pair neutrinos, the muon neutrinos, and the heavy-lepton neutrinos. The electron antineutrinos are suppressed for a while, due to the fact that the electron-neutrino phase space in the core has been filled by the neutrinos from electron capture. Only after these neutrinos from electron capture have been able to diffuse out can the pair processes be successful in producing electron antineutrinos. Eventually, as the electron-capture neutrinos have all diffused out, these electron antineutrinos have a flux equal to that of the electron neutrinos; however, that is not in the initial collapse, but only after several tenths of a second.

Figure 7 shows the trajectories of mass zones as a function of time for a particular model of Wilson's (101), which caused mass ejection. Notice how the zones undergo a hydrodynamic bounce. In this particular model, which had the opacities in the mantle turned up by a factor of 3 to allow ejection, as the bounce material reached its peak and was about to fall back in, it was hit by the neutrino flux and was thus, due to this factor of 3 in the opacity, at the Eddington limit and able to drive off the material before it fell back onto the core. Thus ejection, and a supernova, occurred. If, for some other reason, the neutrino flux could be increased or if the location of the bounce could be slightly changed relative to the neutrino flux, one might also be able to obtain an ejection. Due to the large uncertainties in the equation of state, which determines the strength and location of the bounce, and due to some uncertainties still present in the neutrino transport out of the core, it is clear that the calculations are by no means definitive at the present time. It has been found (65, 101) that by slight changes in the equation of state, well within the range of uncertainty, the hydrodynamic bounce can become sufficiently large that it alone is capable of causing mass ejection. Thus, depending on the equation of state, the bounce may be sufficient to eject matter regardless of the neutrino effects; it may be marginal, so that the

neutrinos are crucial; or it may be so weak that the neutrinos cannot save the day. These equation-of-state uncertainties revolve around the photodisintegration and electron-capture rates for extremely neutron-rich nuclei between the iron-nickel region and free nucleons.

In view of the present status of uncertainty on all of these points, a strong statement cannot be made with regard to whether or not ejection occurs via neutrinos, a bounce, or some other mechanism in the core. However, it does seem reasonable to say that in view of the fact that these stars can produce the heavy elements and seem to be able to produce the cosmic rays and, apparently, definitely have very dense cores that would lend themselves to producing dense remnants like neutron stars or black holes, that this mass range of stars ($M_{\text{CD}} \lesssim M \lesssim M_{\text{pair}}$) is the one of key importance to the supernova problem. However, the actual mechanism whereby the ejection occurs is still a matter of great uncertainty. The weak interaction is obviously of great importance to the whole scenario; however, whether it is the key element in causing an ejection or just an important factor depends on the actual equation of state.

In order to try to observe gravitational collapse, it is important to remember that the neutrinos coming out of the core, whether or not they are able to cause ejection of matter, are still the dominant mechanism by which the energy is lost from the core and they are going to be emitted on a time scale somewhat greater than the collapse time, which in turn is somewhat greater than the free-fall time. In fact, calculations to date seem to indicate that the collapse time is $\gtrsim 5$ times greater than the free-fall time. Thus the time scale over which one would detect the neutrino pulse from gravitational collapse would be of the order of tens of milliseconds, with the duration

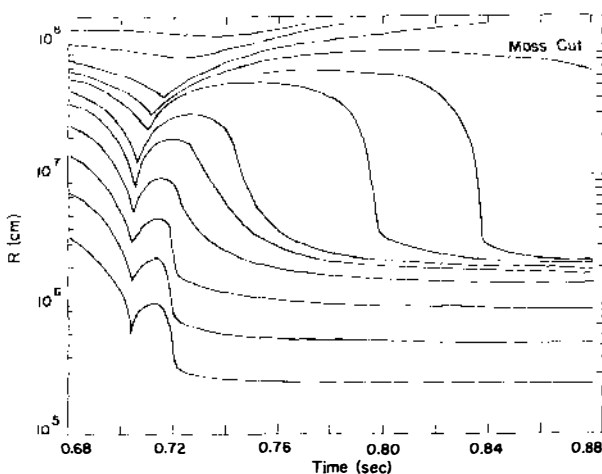


Figure 7 The trajectories of various mass zones in a calculation of Wilson's (101), with the coherent scattering opacity arbitrarily increased by a factor of ~ 3 . The mass ejection occurs when the shocked mantle material is hit by the neutrino flux (compare times with figure 6) before it can fall back on the core.

of the radiation lasting for the amount of time required for the core to settle down to a neutron star, which is the order of tenths of seconds.

Another important point for possible detection is the energy of these neutrinos. The electron-capture neutrinos are going to be emitted at roughly 15–20 MeV. [The bulk of the electron captures take place when the Fermi energy is of the order of 24 MeV (88), and the electron-capture neutrinos have on the average an energy of $\frac{2}{3}$ of the Fermi energy (88).] This makes the neutrinos very low in energy, less than ~ 20 MeV, thus making it very difficult for detectors such as the Deep Underwater Muon and Neutrino Detector (DUMAND) (102) to see extragalactic supernovae. The thermal neutrinos continue to escape after the electron captures cease. The thermal neutrinos are the only ones that produce the easier-to-detect electron antineutrinos, which are inhibited, as we mentioned before, by a factor of 10–100 over the electron neutrinos until well after the initial burst of neutrinos. The thermal neutrinos have an energy characteristic of the temperature at which they are emitted, so as the core continues to heat up, the energy of the thermal neutrinos continues to increase. However, it never seems to reach an extremely high temperature in the core; that is, the energy of the thermal neutrinos never seems to get greater than a few tens of MeVs, again indicating that the detection of these neutrinos would be extremely difficult.

3.2 *Summary of Uncertainties Regarding the Collapse*

Understanding core collapse has proved difficult in the past; one factor is the problem of understanding the presupernova evolution; that is, the relative composition of the different zones in the star and the density profile of the core and the material just outside of the core. This can be strongly influenced by convection calculations.

We can summarize the problems in understanding core collapse as follows:

1. The equation of state, including the composition as the core collapses, in particular taking into account the rate of electron captures on complex nuclei and the rate of photodisintegration of these complex nuclei. This can make very large changes since, if many heavy nuclei are still present in the central core, they will significantly increase the neutrino opacity in the central core and thus significantly change the equation of state and the pressure balance in the central core. A very important point here with regard to the equation of state is the density at which the core bounce takes place and the temperature of the material at the core bounce. This is extremely sensitive to the equation of state and the composition.

2. The second problem is the weak-interaction theory—not just the strength of the isoscalar component of the neutral current that determines the strength of the coherent scattering opacity, but also the number of types of neutrinos that exist; that is, the number of heavy leptons and neutrinos associated with them, since we have seen that the more heavy leptons with their associated neutrinos, the greater the flux of neutrinos coming out of the core that can add to exceed the Eddington limit. As for the coherent scattering, the type of neutrino does not matter, so long as it interacts via a neutral current.

3. The third problem involves the neutrino transport. This must include the effect of both neutrino and electron degeneracy and the effect of conservative and non-

conservative scattering to be able to follow the neutrinos coming out of the core in both their flux and energy. As we have noted, the energy spectrum is extremely important due to the sensitive dependence of neutrino cross sections upon the energy of the neutrino.

4. The fourth problem is that of the effects of rotation and magnetic fields. The calculations mentioned so far have been with a spherically symmetric, nonrotating collapsing star. No rotation or magnetic fields have been mentioned. It seems clear that if rotation and magnetic fields are added, one could change the nature of the problem. In fact, in calculations by LeBlanc & Wilson (103) and further work by Meier et al (104), an ejection of a very different nature was obtained due to the magnetic field winding up and ejecting jets from the poles during a collapse. The effects of rotation and magnetic fields on the pressure balance, etc have not been explored in detail in any of the present calculations.

So we see that there are many problems in the present core collapse; however, these problems, although they affect the final stages of the collapse and whether or not mass is ejected, as far as neutrino detection is concerned, are of relatively little importance since the main fact that neutrinos are emitted in copious numbers from collapse and that the neutrinos emitted are relatively low in energy seems to be relatively insensitive to these details.

Let us summarize. The present models for a nonrotating star always yield a collapsing core greater than $1.1 M_{\odot}$. If it is necessary to eject $\sim \frac{1}{10} M_{\odot}$ of iron in order to explain the iron abundances that are in the galaxy, and it seems that neutrinos may eject that sort of mass, we would end up with a collapsing core mass of about $1.3 M_{\odot}$, which would be quite consistent with the observed masses of neutron stars (66). We also see that if we have a mass average of the ejecta of all of these massive objects, we would get a good fit to the carbon-to-iron abundances in the solar system and we also see that if we have a slightly different way of averaging the supernovae, we could get a reasonable fit to the cosmic-ray abundances and could even make predictions with regard to the cosmic-ray isotopic composition, which has yet to be determined.

We have seen that the bulk of the binding energy is released as neutrinos in these stars. In fact, it has been shown that even nonspherical perturbations are more rapidly damped out via neutrino radiation than via gravitational radiation (105, 106). However, it seems to be difficult to carry away significant angular momentum from the core via neutrino radiation (107, 117). Although neutrinos do remove some angular momentum (96, 107), they also remove energy, and thus the net change in the angular momentum per unit mass is small (117).

We do have some problems with the core. One very serious one is the conditions at the mass cut, that is, the boundary between the matter that implodes to make the neutron star or black hole and the matter that is ejected. Are these conditions at the mass cut similar to those that give the *r*-process? The *r*-process is a nucleosynthetic process thought to be responsible for the neutron-rich heavy elements. The standard scenario places the *r*-process at this boundary between the matter that falls in and the matter that is ejected (39), but until we understand the conditions at the mass cut, we cannot verify this scenario. Another very important question is whether the location of the mass cut is sensitive to the mass of the star. For example, is it possible that a

more massive star, say 40 or 50 $\odot$, might have a collapsing core that would be somewhat greater in mass than the lower-mass stars? In fact, in the stellar-evolution calculations, these extremely massive stars tend to have cores slightly over the Chandrasekhar limit. More massive stars still have the silicon shell burning very rapidly and at very high density, so that it, too, could form part of the collapsing core, whereas the lower-mass stars near 10–15 $M_\odot$ tend to have a core much closer to the 1.4- $M_\odot$ core and evolve in such a way that there is a very thin silicon shell so that it could not significantly increase the mass of the core. Thus the more massive cores may be over the Oppenheimer-Volkoff limit and would collapse to form black holes, whereas the less massive ones would simply make neutron stars. Thus, this question of which-mass stars make black holes and which make neutron stars is one of great uncertainty at the present time and one that will depend on understanding how the mass cut varies with the mass of the star. Another possibility may be that all of these nonrotating, noninteracting stars would yield neutron stars, and that only by putting some other property into the collapse, such as rotation, magnetic fields, or binary interaction, could one get a black hole. The only black holes noted today are in binary systems. (Of course, it would also be very difficult to observe a black hole that was not in a binary system.) On this point, one other comment that should be noted is that, although it seems to be an extremely delicate matter as to whether or not one gets mass ejection in the collapsing core, this is not as bad as it might first seem, since it may mean that a very delicate change one way or another could affect whether a collapse would yield a black hole or a neutron star and ejection. Since we probably see both neutron stars and black holes, the need for some sort of delicate shift from one to the other may be correct.

Let us reiterate that one problem that really needs to be explored is the dependence of the mass of the core on rotation. Where does rotation enter into this thing? How is angular momentum lost from a collapsing star? We know that massive stars on the main sequence are rotating relatively rapidly. Do the cores form while rapidly rotating or have they lost their angular momentum prior to their formation? We do know that all white dwarfs for which rotational velocities are known are slowly rotating. We also know that all pulsars are observed to be slowly rotating. However, that does not answer the questions of whether or not the pulsars formed are rapidly rotating and what the initial velocity of rotation is of a forming neutron star.

In conclusion, we have seen that weak-interaction processes are extremely important to the later stages of stellar evolution, in particular the final gravitational collapse to form a neutron star or a black hole where the bulk of the binding energy is emitted as neutrinos. In fact, the weak neutral current may even be the mechanism that enables the core to collapse while ejecting the outer regions in a supernova explosion. However, until a better understanding of the equation of state of the collapsing matter is obtained, no strong conclusion can be made.

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